

Sensorless Control of Induction Motors Using Cascaded Kalman Filters: Observability and Robustness

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Abstract: *The control of induction motors without mechanical sensors has gained significant attention due to its cost-effectiveness and robustness in harsh environments. This paper evaluates a novel observer-based approach utilizing a cascade of Kalman filters for estimating rotor speed and flux using only stator current and voltage measurements. Given the nonlinear nature of the system, the separation principle is not guaranteed, making observability and robustness key concerns. Two control strategies are examined: singular perturbation-based control and feedback linearization. A stability analysis using Lyapunov methods is performed to ensure robustness against estimation errors. The effectiveness of the proposed approach is demonstrated through numerical simulations.*

Contents

1	Introduction and Context	2
1.1	Challenges in Sensorless State Estimation	3
1.2	Recherche Objectives	3
1.3	Organization of the Article	3
2	Induction Motor Modeling	4
2.1	Motor Parameters and Simulation Conditions	4
2.2	Mathematical representation in a general xy reference frame	4
2.3	Mathematical representation in $\alpha - \beta$ reference frame	5
2.4	The equations in the DQ reference frame	5

3	Control Strategies Based on State Feedback	7
3.1	Singular Perturbation-Based Control	7
3.1.1	Control Equations	7
3.1.2	Justification of the Controller Selection and and Stability Analysis . . .	8
3.1.3	Implementation and Tuning	9
3.1.4	Simulation results	10
3.2	Feedback Linearization Control	11
3.2.1	Control Equations	11
3.2.2	Justification of the Controller Selection and and Stability Analysis . . .	13
3.2.3	Implementation and Tuning	14
3.2.4	Simulation results	16
4	Observer	17
4.1	Motor equations	17
4.2	Cascade Kalman filters	18
4.2.1	Observability of flux subsystem	20
4.2.2	Observability of velocity subsystem	21
4.3	Uniform Complete Controllability	22
4.4	Exponential Stability	24
5	Output Feedback Control and Robustness	25
5.1	Observer and Controller Interaction	25
5.2	Stability Validation Using Lyapunov's Method	26
5.3	Application to singular perturbations control	28
5.3.1	Sensitivity results to perturbations	30
5.3.2	Sensitivity results to modeling errors	31
5.4	Application to feedback linearization control	32
5.4.1	Sensitivity results to perturbations	33
5.4.2	Sensitivity results to modeling errors	34
6	Conclusions	35

1 Introduction and Context

Induction motors play a fundamental role in industrial and transportation systems due to their robustness, efficiency, and low maintenance costs. They are extensively used in applications ranging from manufacturing processes to household appliances. Traditional control strategies rely on mechanical sensors, such as encoders and resolvers. However, these sensors present several drawbacks: they increase system costs, are susceptible to mechanical wear and failures, and may be unsuitable for harsh environments with high temperatures, vibrations, or contamination.

As a result, sensorless control strategies have gained considerable attention, as they enable rotor speed and flux estimation using only electrical measurements, such as stator currents and voltages. By eliminating mechanical sensors, these approaches enhance system reliability and reduce maintenance needs. However, the effectiveness of sensorless control heavily depends on the accuracy of the state estimation process.

1.1 Challenges in Sensorless State Estimation

The primary challenge in sensorless control of induction motors lies in the nonlinear nature of their mathematical model. Unlike linear systems, where the separation principle ensures that estimation and control can be designed independently, in nonlinear systems this principle no longer holds. Consequently, designing an observer that guarantees accurate state estimation and ensures overall system stability remains an open research problem.

In [[5]], an novel estimation technique based on Kalman filtering was proposed to improve state reconstruction in nonlinear systems. Specifically, a cascaded Kalman filter observer was introduced to enhance estimation accuracy by leveraging multiple filtering stages. However, the observability of the system under such an approach must be rigorously analyzed, as poor observability can lead to estimation errors that compromise control performance. Furthermore, robustness against modeling uncertainties and external disturbances must be ensured to guarantee reliable operation in practical applications.

1.2 Recherche Objectives

This study evaluates the feasibility of using cascaded Kalman filters for sensorless control of induction motors. The research focuses on two key control strategies: singular perturbation-based control and feedback linearization. The robustness of the proposed methods is assessed through Lyapunov-based stability analysis. The specific objectives of this study are:

- To develop and compare two sensorless control strategies for induction motors.
- To implement a cascaded Kalman filter observer using only stator current and voltage measurements.
- To analyze observability conditions and identify potential limitations in state estimation.
- To evaluate the robustness of the proposed control strategies through Lyapunov-based stability analysis.

1.3 Organization of the Article

This article is organized as follows: Section 2 presents the mathematical modeling of the induction motor, starting with the motor parameters and simulation conditions, followed by mathematical representations in the general xy -reference frame, the $\alpha - \beta$ reference frame, and the dq -reference frame. Section 3 details the proposed control strategies based on state feedback, focusing on singular perturbation-based control and feedback linearization, including control equations, stability analysis, and implementation. Section 4 introduces the observer, specifically the cascaded Kalman filter approach, and includes an observability analysis of the flux and velocity subsystems, as well as an exploration of uniform complete controllability and exponential stability. Section 5 discusses output feedback control implementation, stability validation using Lyapunov's method, and robustness analysis for both singular perturbation and feedback linearization controls, including sensitivity results to perturbations and modeling errors. Finally, Section 6 concludes the paper and suggests future research directions.

2 Induction Motor Modeling

The modeling of the induction motor is essential for the design and evaluation of sensorless control strategies. Accurate motor parameters and well-defined simulation conditions play a critical role in ensuring the reliability and performance of the control algorithms. In this section, the system is first expressed in a general xy rotating frame, where the dynamics are initially described. This representation is then simplified to the stationary $\alpha - \beta$ frame, followed by a transformation into the rotating DQ frame. The DQ frame facilitates control design by aligning the system dynamics with the rotor flux, effectively decoupling flux and torque components.

2.1 Motor Parameters and Simulation Conditions

For the simulation studies conducted in this research, the test motor is characterized by the following nominal parameters. These values were used as a basis for the development and evaluation of the sensorless control strategies.

Table 1: Test motor rated values and parameters

Parameter	Value	Parameter	Value
Rated Power	4 kW	Rotor Inductance, L_r	225 mH
Rated Voltage (RMS)	400 V	Stator Inductance, L_s	225 mH
Rated Frequency	50 Hz	Mutual Inductance, L_m	221 mH
Rated Current	6 A	Rotor Resistance, R_r	0.745 Ω
Rated Torque	30.9 N·m	Stator Resistance, R_s	1.22 Ω
Number of Pole Pairs	2	Motor Moment of Inertia, J_m	0.04 kg·m ²
Nominal Flux	1.2275 Wb	Total Moment of Inertia, J	0.1 kg·m ²

Additionally, maximum admissible peak values for voltages and currents are considered in the analysis, with these values not being sustained for more than 1 millisecond to avoid damage to the system.

Table 2: Maximum admissible peak values

Parameter	Value
Peak Voltage (RMS)	800 V
Peak Current	60 A

In the control strategies analyzed in this work, two degrees of freedom are considered: the motor speed, which is dictated by the application requirements, and the magnetic flux, which is set to its nominal value as a reference for the purpose of this study.

2.2 Mathematical representation in a general xy reference frame

The following standard equations describing the induction motor dynamics are taken from [5], where they are derived based on an energy approach or a microscopic approach under the assumption of no magnetic saturation. These equations are expressed in a general xy reference frame with an arbitrary frame velocity ω_e :

$$\frac{d\psi_s^{xy}}{dt} = u_s^{xy} - \omega_e \mathcal{J} \psi_s^{xy} - R_s i_s^{xy} \quad (1a)$$

$$\frac{d\psi_r^{xy}}{dt} = (\omega_r - \omega_e) \mathcal{J} \psi_r^{xy} - R_r i_r^{xy} \quad (1b)$$

$$\begin{bmatrix} i_s^{xy} \\ i_r^{xy} \end{bmatrix} = \begin{bmatrix} \mathcal{L}_s & \mathcal{L}_m \\ \mathcal{L}_m & \mathcal{L}_r \end{bmatrix}^{-1} \begin{bmatrix} \psi_s^{xy} \\ \psi_r^{xy} \end{bmatrix} \quad (1c)$$

$$\frac{J}{n} \frac{d\omega_r}{dt} = T_{em} - T_l \quad (1d)$$

$$T_{em} = nL_m \langle i_s^{xy}, \mathcal{J} i_r^{xy} \rangle \quad (1e)$$

2.3 Mathematical representation in $\alpha - \beta$ reference frame

By setting the frame velocity to $\omega_e = 0$, the equations simplify to the $\alpha - \beta$ stationary reference frame, where the motor equations are expressed in terms of the stator and rotor flux components, along with the stator currents:

$$\frac{d\psi_s^{\alpha\beta}}{dt} = u_s^{\alpha\beta} - R_s i_s^{\alpha\beta} \quad (2a)$$

$$\frac{d\psi_r^{\alpha\beta}}{dt} = \omega_r \mathcal{J} \psi_r^{\alpha\beta} - R_r i_r^{\alpha\beta} \quad (2b)$$

$$\begin{bmatrix} i_s^{\alpha\beta} \\ i_r^{\alpha\beta} \end{bmatrix} = \begin{bmatrix} \mathcal{L}_s & \mathcal{L}_m \\ \mathcal{L}_m & \mathcal{L}_r \end{bmatrix}^{-1} \begin{bmatrix} \psi_s^{\alpha\beta} \\ \psi_r^{\alpha\beta} \end{bmatrix} \quad (2c)$$

$$\frac{J}{n} \frac{d\omega_r}{dt} = T_{em} - T_l \quad (2d)$$

$$T_{em} = nL_m \langle i_s^{\alpha\beta}, \mathcal{J} i_r^{\alpha\beta} \rangle \quad (2e)$$

where $\psi_s^{\alpha\beta} = (\psi_s^\alpha, \psi_s^\beta) \in \mathbb{R}^2$, $i_s^{\alpha\beta} = (i_s^\alpha, i_s^\beta) \in \mathbb{R}^2$, $\psi_r^{\alpha\beta} = (\psi_r^\alpha, \psi_r^\beta) \in \mathbb{R}^2$, $i_r^{\alpha\beta} = (i_r^\alpha, i_r^\beta) \in \mathbb{R}^2$, $\mathcal{J} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, $\mathcal{L}_s = L_s \mathbb{I}_2$, $\mathcal{L}_m = L_m \mathbb{I}_2$, $\mathcal{L}_r = L_r \mathbb{I}_2$ and $\langle a, b \rangle := a^T b$

The inductances are multiplied by the 2×2 identity matrix to ensure the dimensions of the matrices are consistent.

2.4 The equations in the DQ reference frame

To simplify the analysis and controller design, the model is transformed into the rotating DQ reference frame, which is aligned with the rotor flux. This transformation is achieved through the Clarke and Park transformations, enabling the decoupling of the flux and torque components. This decoupling simplifies the control design by allowing independent regulation of the rotor flux magnitude (via the D-axis current) and the electromagnetic torque (via the Q-axis current). The rotation of the reference frame is defined by the angle of the rotor flux

vector, ensuring dynamic alignment with the rotor's magnetic field. This alignment allows for direct control of the magnetic flux magnitude, providing a more intuitive and effective approach to managing the motor's performance.

In this framework, the rotor flux is expressed in polar form, which facilitates the analysis and control of the induction motor. The polar representation of the rotor flux is given by:

$$\rho = \tan^{-1} \left(\frac{\psi_r^\beta}{\psi_r^\alpha} \right) \quad (3a)$$

$$\Psi = \sqrt{(\psi_r^\alpha)^2 + (\psi_r^\beta)^2}. \quad (3b)$$

Thus, the stator currents and voltages in the DQ frame are given by:

$$\begin{bmatrix} i_s^D \\ i_s^Q \end{bmatrix} = \begin{bmatrix} \cos(\rho) & \sin(\rho) \\ -\sin(\rho) & \cos(\rho) \end{bmatrix} \begin{bmatrix} i_s^\alpha \\ i_s^\beta \end{bmatrix} \quad (4a)$$

$$\begin{bmatrix} u_s^D \\ u_s^Q \end{bmatrix} = \begin{bmatrix} \cos(\rho) & \sin(\rho) \\ -\sin(\rho) & \cos(\rho) \end{bmatrix} \begin{bmatrix} u_s^\alpha \\ u_s^\beta \end{bmatrix} \quad (4b)$$

This representation allows the decoupling of flux and torque dynamics, facilitating the design of control strategies:

$$\frac{L_r}{R_r} \frac{d\Psi}{dt} = -\Psi + L_m i_s^D \quad (5a)$$

$$\frac{J}{n} \frac{d\omega_r}{dt} = n \frac{L_m}{L_r} \Psi i_s^Q - T_l \quad (5b)$$

$$L_f \frac{di_s^D}{dt} = - \left(R_s + \frac{L_m^2 R_r}{L_r^2} \right) i_s^D + \frac{L_m R_r}{L_r^2} \Psi + L_f L_m \frac{R_r}{L_r} \frac{i_s^{Q^2}}{\Psi} + L_f \omega_r i_s^Q + u_s^D \quad (5c)$$

$$L_f \frac{di_s^Q}{dt} = - \left(R_s + \frac{L_m^2 R_r}{L_r^2} \right) i_s^Q - \frac{L_m}{L_r} \omega_r \Psi - L_f \omega_r i_s^D - L_f L_m \frac{R_r}{L_r} \frac{i_s^D i_s^Q}{\Psi} + u_s^Q \quad (5d)$$

It is possible to redefine the model using a normalized flux magnitude, expressed as $\Phi = \frac{L_m}{L_r} \Psi$, to simplify the analysis. Furthermore, the reliance of the control design on unknown parameters can be minimized to the following four variables:

$$R_{req} = R_r \frac{L_m^2}{L_r^2}, \quad T_r^{-1} = \frac{R_r}{L_r}, \quad R_s, \quad L_f = L_s - \frac{L_m^2}{L_r} \quad (6)$$

With this change of variables, the equations in 5 are rewritten as:

$$\frac{d\Phi}{dt} = -\frac{1}{T_r} \Phi + R_{req} i_s^D \quad (7a)$$

$$\frac{J}{n} \frac{d\omega_r}{dt} = n \Phi i_s^Q - T_l \quad (7b)$$

$$L_f \frac{di_s^D}{dt} = - (R_s + R_{req}) i_s^D + \frac{1}{T_r} \Phi + L_f R_{req} \frac{i_s^{Q^2}}{\Phi} + L_f \omega_r i_s^Q + u_s^D \quad (7c)$$

$$L_f \frac{di_s^Q}{dt} = - (R_s + R_{req}) i_s^Q - \omega_r \Phi - L_f \omega_r i_s^D - L_f R_{req} \frac{i_s^D i_s^Q}{\Phi} + u_s^Q \quad (7d)$$

This formulation highlights the influence of the system inputs on the state variables, providing a solid foundation for controller design. By transforming the model into the DQ reference frame, the dynamics of the induction motor are decoupled into two independent components: the D-axis, which governs the rotor flux magnitude, and the Q-axis, which controls the electromagnetic torque. This decoupling simplifies the control strategy, enabling independent regulation of flux and torque through the D-axis and Q-axis currents, respectively. As a result, the design of control laws becomes more intuitive and systematic.

3 Control Strategies Based on State Feedback

3.1 Singular Perturbation-Based Control

The singular perturbation approach exploits the natural time-scale separation in the induction motor dynamics. The rotor flux magnitude (Ψ) and rotor angular velocity (ω_r) evolve more slowly compared to the stator current dynamics. By leveraging this separation, the control strategy independently regulates Ψ and ω_r by defining appropriate reference currents. Assuming that the stator currents can accurately track their references with a sufficiently fast response, the desired behavior of Ψ and ω_r can be achieved.

3.1.1 Control Equations

To implement this, a two-level control structure is employed:

- **Speed Control Loop:** Operating at a slower time scale, the speed control loop establishes both the reference quadrature current (i_s^Q) and the reference direct current (i_s^D) required to drive the rotor flux magnitude (Φ) and rotor speed (ω_r) to their desired equilibrium values while ensuring stability:

$$i_{s,\text{ref}}^D = \left(\frac{1}{R_{\text{req}}} \right) \left(\frac{1}{T_r} \Phi_{\text{ref}} - \lambda_1 (\Phi - \Phi_{\text{ref}}) \right) \quad (8a)$$

$$i_{s,\text{ref}}^Q = \frac{1}{n\Phi} (T_l - \lambda_2 (\omega_r - \omega_{r,\text{ref}})) \quad (8b)$$

- **Current Control Loop:** To ensure that the actual stator currents follow their reference values rapidly, a proportional-integral (PI) controller is implemented. The controller employs sufficiently high gains to dominate the nonlinear dynamics and reinforce the time-scale separation strategy:

$$u_s^D = -K_P^D (i_s^D - i_{s,\text{ref}}^D) - \eta_D \quad (9a)$$

$$\frac{d\eta_D}{dt} = K_I^D (i_s^D - i_{s,\text{ref}}^D) \quad (9b)$$

$$u_s^Q = -K_P^Q (i_s^Q - i_{s,\text{ref}}^Q) - \eta_Q \quad (9c)$$

$$\frac{d\eta_Q}{dt} = K_I^Q (i_s^Q - i_{s,\text{ref}}^Q) \quad (9d)$$

3.1.2 Justification of the Controller Selection and and Stability Analysis

The selection of the proposed controller is based on the dynamic behavior of the rotor angular velocity (ω_r) and the rotor flux magnitude (Ψ). Assuming the load torque (T_l) is constant and known, the dynamics of ω_r depend solely on the quadrature current (i_s^Q) and the magnitude of the rotor flux (Ψ). Notably, Ψ exhibits an independent dynamic behavior influenced by itself and the direct current (i_s^D). This decoupling allows us to set the reference currents independently, which enables the desired dynamics for both Ψ and ω_r to be achieved.

On the assumption that the reference values for the direct current (i_s^D) and the quadrature current (i_s^Q) can be accurately established, or at least follow their references with dynamics significantly faster than those of the flux (Ψ) and rotor velocity (ω_r), the dynamics of both Ψ and ω_r can be effectively directed toward the desired behavior. The expressions for these reference currents defined in 8 and will be used to achieve the intended dynamics.

$$\frac{d\Phi}{dt} = -\frac{1}{T_r}\Phi + R_{req}i_{s,\text{ref}}^D \quad (10a)$$

$$\frac{d\Phi}{dt} = -\frac{1}{T_r}\Phi + R_{req} \left(\frac{1}{R_{req}} \right) \left(\frac{1}{T_r}\Phi_{ref} - \lambda_1 (\Phi - \Phi_{ref}) \right) \quad (10b)$$

$$\frac{d\Phi}{dt} = -\left(\frac{1}{T_r} + \lambda_1 \right) (\Phi - \Phi_{ref}) \quad (10c)$$

$$\frac{d(\Phi - \Phi_{ref})}{dt} = -\left(\frac{1}{T_r} + \lambda_1 \right) (\Phi - \Phi_{ref}) - \dot{\Phi}_{ref} \quad (10d)$$

$$\frac{J}{n} \frac{d\omega_r}{dt} = n\Phi i_{s,\text{ref}}^Q - T_l \quad (11a)$$

$$\frac{J}{n} \frac{d\omega_r}{dt} = n\Phi \frac{1}{n\Phi} (T_l - \lambda_2 (\omega_r - \omega_{r,\text{ref}})) - T_l \quad (11b)$$

$$\frac{d\omega_r}{dt} = -\frac{n\lambda_2}{J} (\omega_r - \omega_{r,\text{ref}}) \quad (11c)$$

$$\frac{d(\omega_r - \omega_{r,\text{ref}})}{dt} = -\frac{n\lambda_2}{J} (\omega_r - \omega_{r,\text{ref}}) - \dot{\omega}_{r,\text{ref}} \quad (11d)$$

By substituting the reference currents into the governing equations of Φ and ω_r (see equations 10 and 11), the dynamics can be reformulated. Considering the rotor flux magnitude (Φ) and the rotor angular velocity (ω_r) as constant desired values, the substitution leads to expressions of the following form:

$$\frac{de_\Phi}{dt} = -\lambda_\Phi e_\Phi \quad (12a)$$

$$\frac{de_{\omega_r}}{dt} = -\lambda_{\omega_r} e_{\omega_r} \quad (12b)$$

where $e_\Phi = \Phi - \Phi_{\text{ref}}$ and $e_{\omega_r} = \omega_r - \omega_{r,\text{ref}}$. The constants $\lambda_\Phi = \frac{1}{T_r} + \lambda_1$ and $\lambda_{\omega_r} = \frac{n_p\lambda_2}{J}$ represent the rates at which the errors e_Φ and e_{ω_r} decay.

This structure corresponds to a first-order linear differential equation, which is globally exponentially stable. The exponential decay ensures that the errors e_Φ and e_{ω_r} converge to

zero over time, provided the desired references are constant. This stability confirms that achieving the reference currents ensures precise control of both rotor flux magnitude and angular velocity.

According to Theorem 9.4 in [2], considering the system as a singular perturbation problem, the reference currents act as isolated roots defining the equilibrium of the fast subsystem. If the origin of the reduced model, obtained by substituting these reference currents into the slow dynamics, is exponentially stable, as demonstrated in 11, 10 and 12, then the full singularly perturbed system has a unique solution that asymptotically follows the reduced model.

Furthermore, the stability of the boundary layer problem, which corresponds to the fast current dynamics when the slow variables are held constant, must also be ensured. In this case, the PI control with sufficiently high gains enforces rapid convergence of the currents to their reference values, thereby stabilizing the fast dynamics and reinforcing time-scale separation.

This theoretical foundation justifies the proposed control strategy, ensuring that the overall system follows the desired trajectory and remains stable under the singular perturbation framework.

3.1.3 Implementation and Tuning

The tuning of the controllers is based on transient response and stability criteria to ensure that the currents follow their reference values rapidly and in a synchronized manner. Initially, the current controller parameters (K_P, K_I) are determined by considering a settling time t_s , derived from the open-loop response, with a damping ratio $\zeta = 0.7$. This criterion ensures that the system has a critically damped response without excessive oscillations.

$$\begin{bmatrix} \frac{di_s^D}{dt} \\ \frac{d\eta_D^D}{dt} \end{bmatrix} = \begin{bmatrix} -\frac{\left(R_s + \frac{L_m^2 R_r}{L_r^2}\right)}{L_f} - \frac{K_P^D}{L_f} & -\frac{1}{L_f} \\ K_I^D & 0 \end{bmatrix} \begin{bmatrix} i_s^D \\ \eta_D^D \end{bmatrix} + \begin{bmatrix} b_1 \\ -K_I^D i_{s,\text{ref}}^D \end{bmatrix} \quad (13a)$$

$$\begin{bmatrix} \frac{di_s^Q}{dt} \\ \frac{d\eta_Q^Q}{dt} \end{bmatrix} = \begin{bmatrix} -\frac{\left(R_s + \frac{L_m^2 R_r}{L_r^2}\right)}{L_f} - \frac{K_P^Q}{L_f} & -\frac{1}{L_f} \\ K_I^Q & 0 \end{bmatrix} \begin{bmatrix} i_s^Q \\ \eta_Q^Q \end{bmatrix} + \begin{bmatrix} b_2 \\ -K_I^Q i_{s,\text{ref}}^Q \end{bmatrix} \quad (13b)$$

$$P_1(s) = s^2 + s \frac{\left(R_s + \frac{L_m^2 R_r}{L_r^2} + K_P^D\right)}{L_f} + \frac{K_I^D}{L_f} \quad (13c)$$

$$P_2(s) = s^2 + s \frac{\left(R_s + \frac{L_m^2 R_r}{L_r^2} + K_P^Q\right)}{L_f} + \frac{K_I^Q}{L_f} \quad (13d)$$

From the characteristic polynomial equations 13c and 13d, derived from 13a and 13b, the same settling time can be selected for both equations to ensure synchronized convergence of i_s^D to $i_{s,\text{ref}}^D$ and i_s^Q to $i_{s,\text{ref}}^Q$, preventing any phase mismatch in the system's response. The decision to use a settling time of $t_s=0.001\text{s}$ was made based on the open-loop stability time of $t_s=0.005\text{s}$, with a damping ratio $\zeta = 0.7$, ensuring a sufficiently fast and synchronized response. Adjustments to the proportional (K_P) and (K_I) integral gains will be made accordingly to maintain synchronization following the 2% criterion of stability [3]

$$t_s = \frac{4}{\zeta\omega_0} \quad (14a)$$

$$\omega_0 = \sqrt{\frac{K_I}{L_f}} \quad (14b)$$

$$\zeta = \frac{\left(R_s + \frac{L_m^2 R_r}{L_r^2} + K_P\right)}{2\sqrt{L_f K_I}} \quad (14c)$$

Derived from the preceding equations (14a, 14b, and 14c), the calculated proportional and integral gains are $K_P = 78.28$ and $K_I = 326\ 530$.

Subsequently, the speed loop parameters λ_1 and λ_2 are selected. given that higher values of λ_1 and λ_2 lead to faster convergence, the largest possible values are chosen while respecting the motor's physical limitations. These constraints are based on the nominal values and permissible maximum limits of the motor, as shown in Table 2.

Finally, the values for λ_1 and λ_2 are selected as 0.7, which were empirically obtained from simulation results to ensure a balanced performance between fast convergence and the avoidance of excessive stress on the motor.

3.1.4 Simulation results

The simulation is divided into three time intervals to analyze the system's response under different conditions:

- **Interval 1** ($t = 0$ to $t = 0.5$ s): The system is initialized at equilibrium, where all state variables are set to their nominal values.
- **Interval 2** ($t = 0.5$ to $t = 1.0$ s): A change in the reference speed is introduced to evaluate the controller's tracking performance.
- **Interval 3** ($t = 1.0$ to $t = 1.5$ s): At $t = 1.0$ s, an abrupt increase of 50% in the external torque is applied.

Where the system operates under the assumption that all motor states and external torque are perfectly known.

The simulation results demonstrate the effectiveness of the singular perturbation-based control under ideal conditions. Initially, the system remains stable at its nominal equilibrium point, as expected. When the reference speed is changed, the system follows the new reference (red curve) in an overdamped manner, reaching steady-state in approximately 0.25 s.

Furthermore, after the external torque disturbance, no significant deviation is observed in the system's response. This indicates that knowing the exact disturbance value allows the control signal to adjust immediately and appropriately.

The absolute errors at the end of both the second and third intervals are on the order of 10^{-5} , which is negligible compared to the relatively high motor speeds. Additionally, reducing numerical tolerances in the simulation further decreases these errors, suggesting that they are primarily due to numerical precision limitations of the software.

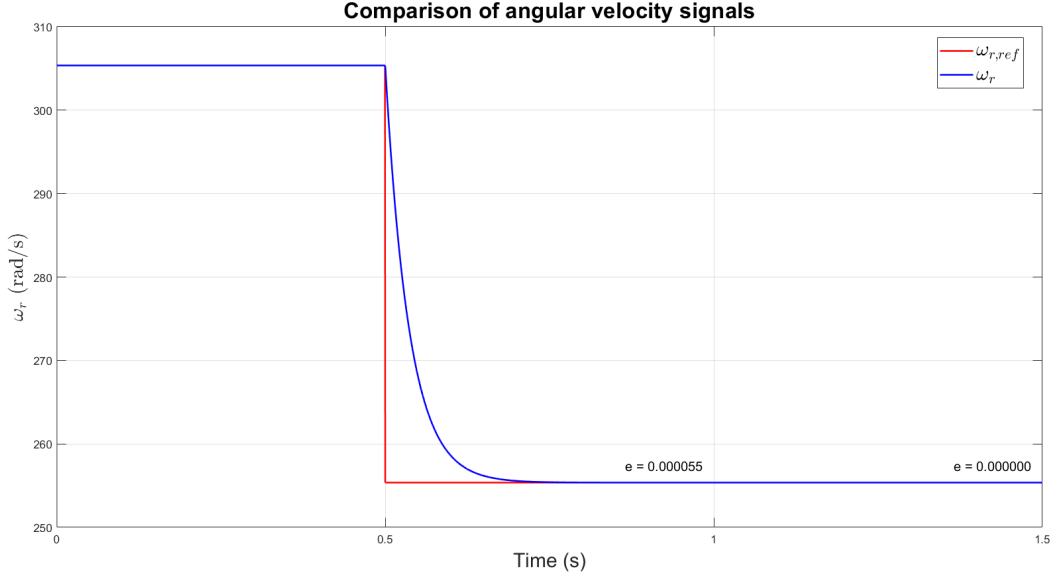


Figure 1: System response using singular perturbation-based control, assuming full state measurement and known external torque.

3.2 Feedback Linearization Control

Feedback linearization transforms the nonlinear dynamics of the induction motor into a linear system by applying an appropriate nonlinear control law. This approach cancels the inherent nonlinearities of the motor model, enabling the use of standard linear control techniques. By defining an appropriate coordinate transformation and control input, the motor's behavior can be shaped to follow desired trajectories with improved performance and stability. This strategy allows precise control of the flux and torque, making it suitable for sensorless control applications.

3.2.1 Control Equations

Flux control through ν_D The flux dynamics are given by:

$$\frac{d\Phi}{dt} = -\frac{1}{T_r}\Phi + R_{req}i_s^D \quad (15a)$$

$$L_f \frac{di_s^D}{dt} = \frac{1}{T_r}\Phi - (R_s + R_{req})i_s^D + L_f R_{req} \frac{i_s^{Q^2}}{\Phi} + L_f \omega_r i_s^Q + u_s^D \quad (15b)$$

This system can be rewritten in a linear state-space form as:

$$\begin{bmatrix} \dot{\Phi} \\ \dot{i}_s^D \end{bmatrix} = \begin{bmatrix} -\frac{1}{T_r} & R_{req} \\ \frac{1}{T_r L_f} & -\frac{R_s + R_{req}}{L_f} \end{bmatrix} \begin{bmatrix} \Phi \\ i_s^D \end{bmatrix} + \begin{bmatrix} 0 \\ \nu_D \end{bmatrix} \quad (16)$$

where the control input is defined as:

$$\nu_D = R_{req} \frac{i_s^{Q^2}}{\Phi} + \omega_r i_s^Q + \frac{1}{L_f} u_s^D. \quad (17)$$

At equilibrium, the flux and direct current components satisfy:

$$\Phi = \Phi_{ref} \quad (18a)$$

$$i_{s,\text{ref}}^D = \frac{1}{R_{req}T_r} \Phi_{ref} \quad (18b)$$

Substituting these into the system dynamics, the equilibrium control input is:

$$\nu_{D,ref} = \frac{R_s}{R_{req}T_r} \frac{\Phi_{ref}}{L_f}. \quad (19)$$

To ensure the system follows the desired reference, a proportional feedback law is applied:

$$\nu_D = -k_{D1}(\Phi - \Phi_{ref}) - k_{D2}(i_s^D - i_{s,\text{ref}}^D) + \nu_{D,ref}. \quad (20)$$

Solving for the actual control input:

$$u_s^D = L_f \left(-R_{req} \frac{i_s^{Q^2}}{\Phi} - \omega_r i_s^Q - k_{D1}(\Phi - \Phi_{ref}) - k_{D2}(i_s^D - i_{s,\text{ref}}^D) + \nu_{D,ref} \right). \quad (21)$$

Speed control through ν_Q To facilitate control design, the auxiliary variable $\tau = \Phi i_s^Q$ is introduced, representing the electromagnetic torque-producing term. The system dynamics in terms of τ are:

$$\frac{J}{n} \frac{d\omega_r}{dt} = n\tau - T_l \quad (22a)$$

$$L_f \frac{d\tau}{dt} = - \left(R_s + R_{req} + \frac{L_f}{T_r} \right) \tau - \Phi(\Phi + L_f i_s^D) \omega_r + \Phi u_s^Q \quad (22b)$$

This system can be rewritten in a linear state-space form as:

$$\begin{bmatrix} \dot{\omega}_r \\ \dot{\tau} \end{bmatrix} = \begin{bmatrix} 0 & \frac{n^2}{J} \\ 0 & -\frac{R_q}{L_f} \end{bmatrix} \begin{bmatrix} \omega_r \\ \tau \end{bmatrix} + \begin{bmatrix} -\frac{n}{J} T_l \\ \nu_Q \end{bmatrix} \quad (23)$$

where:

$$R_q = R_s + R_{req} + \frac{L_f}{T_r}, \quad \nu_Q = -\frac{\Phi}{L_f}(\Phi + L_f i_s^D) \omega_r + \frac{\Phi}{L_f} u_s^Q. \quad (24)$$

At equilibrium, the system satisfies:

$$\omega_r = \omega_{r,\text{ref}} \quad (25a)$$

$$\tau_{ref} = \frac{T_l}{n} \quad (25b)$$

which leads to the equilibrium control input:

$$\nu_{Q,ref} = R_q \frac{T_l}{n L_f}. \quad (26)$$

Applying a proportional feedback law:

$$\nu_Q = -k_{Q1}(\omega_r - \omega_{r,\text{ref}}) - k_{Q2}(\tau - \tau_{ref}) + \nu_{Q,ref}. \quad (27)$$

Solving for the control input:

$$u_s^Q = (\Phi + L_f i_s^D) \omega_r + \frac{L_f}{\Phi} (-k_{Q1}(\omega_r - \omega_{r,\text{ref}}) - k_{Q2}(\tau - \tau_{ref}) + \nu_{Q,ref}). \quad (28)$$

3.2.2 Justification of the Controller Selection and and Stability Analysis

By formulating the system dynamics in terms of error variables and analyzing the corresponding matrices, conditions for stability are established. The analysis covers both flux and current dynamics, as well as speed behavior, showing that the proposed control strategy ensures a stable response.

$$\frac{d\Phi}{dt} = -\frac{1}{T_r}\Phi + R_{req}\iota_s^D \quad (29a)$$

$$\frac{d\Phi}{dt} = -\frac{1}{T_r}\Phi + R_{req}\iota_s^D + \frac{1}{T_r}\Phi_{ref} - R_{req}\iota_{s,ref}^D \quad (29b)$$

$$\frac{d\Phi}{dt} = -\frac{1}{T_r}(\Phi - \Phi_{ref}) + R_{req}(\iota_s^D - \iota_{s,ref}^D) \quad (29c)$$

$$\frac{d\iota_s^D}{dt} = \frac{1}{T_r} \frac{\Phi}{L_f} - \frac{R_s + R_{req}}{L_f} \iota_s^D + \nu_D \quad (30a)$$

$$\frac{d\iota_s^D}{dt} = \frac{1}{T_r} \frac{\Phi}{L_f} - \frac{R_s + R_{req}}{L_f} \iota_s^D - k_{D1}(\Phi - \Phi_{ref}) - k_{D2}(\iota_s^D - \iota_{s,ref}^D) + \nu_{D,ref} \quad (30b)$$

$$\frac{d\iota_s^D}{dt} = \frac{1}{T_r L_f}(\Phi - \Phi_{ref}) - \frac{R_s + R_{req}}{L_f}(\iota_s^D - \iota_{s,ref}^D) - k_{D1}(\Phi - \Phi_{ref}) - k_{D2}(\iota_s^D - \iota_{s,ref}^D) \quad (30c)$$

Considering the following errors, $e_\Phi = \Phi - \Phi_{ref}$ and $e_{\iota_s^D} = \iota_s^D - \iota_{s,ref}^D$ and assuming that the reference values for flux and current are constant, meaning their derivatives are zero, we obtain the following expressions:

$$\frac{de_\Phi}{dt} = -\frac{1}{T_r}e_\Phi + R_{req}e_{\iota_s^D} \quad (31a)$$

$$\frac{de_{\iota_s^D}}{dt} = \frac{1}{T_r L_f}e_\Phi - \frac{R_s + R_{req}}{L_f}e_{\iota_s^D} - k_{D1}e_\Phi - k_{D2}e_{\iota_s^D} \quad (31b)$$

$$\begin{bmatrix} \dot{e}_\Phi \\ \dot{e}_{\iota_s^D} \end{bmatrix} = \underbrace{\begin{bmatrix} -\frac{1}{T_r} & R_{req} \\ \frac{1}{T_r L_f} - k_{D1} & -\frac{R_s + R_{req}}{L_f} - k_{D2} \end{bmatrix}}_M \begin{bmatrix} e_\Phi \\ e_{\iota_s^D} \end{bmatrix} \quad (31c)$$

$$tr(M) = -\frac{1}{T_r} - \frac{R_s + R_{req}}{L_f} - k_{D2} \quad (31d)$$

$$det(M) = \frac{R_s}{T_r L_f} + \frac{k_{D2}}{T_r} + R_{req}k_{D1} \quad (31e)$$

It is found that for all $k_{D1} \geq 0$ and $k_{D2} \geq 0$, the following conditions hold:

$$tr(M) < 0 \quad \text{and} \quad det(M) > 0,$$

indicating that M is a Hurwitz matrix.

Analogously, we analyze the speed dynamics following the same approach.

$$\frac{d\omega_r}{dt} = \frac{n^2}{J}\tau - \frac{n}{J}T_l \quad (32a)$$

$$\frac{d\omega_r}{dt} = \frac{n^2}{J}\tau - \frac{n}{J}T_l + \frac{n^2}{J}\frac{T_l}{n} - \frac{n^2}{J}\tau_{ref} \quad (32b)$$

$$\frac{d\omega_r}{dt} = \frac{n^2}{J}(\tau - \tau_{ref}) \quad (32c)$$

$$\frac{d\tau}{dt} = -\frac{R_q}{L_f}\tau + \nu_Q \quad (33a)$$

$$\frac{d\tau}{dt} = -\frac{R_q}{L_f}\tau - k_{Q1}(\omega_r - \omega_{r,ref}) - k_{Q2}(\tau - \tau_{ref}) + \nu_{Q,ref} \quad (33b)$$

$$\frac{d\tau}{dt} = -\frac{R_q}{L_f}(\tau - \tau_{ref}) - k_{Q1}(\omega_r - \omega_{r,ref}) - k_{Q2}(\tau - \tau_{ref}) \quad (33c)$$

Considering the following errors, $e_{\omega_r} = \omega_r - \omega_{r,ref}$ and $e_\tau = \tau - \tau_{ref}$ and assuming that the reference values for speed, flux and current, and therefore τ , are constant, meaning their derivatives are zero, we obtain the following expressions:

$$\frac{de_{\omega_r}}{dt} = \frac{n^2}{J}e_\tau \quad (34a)$$

$$\frac{de_\tau}{dt} = -\frac{R_q}{L_f}e_\tau - k_{Q1}e_{\omega_r} - k_{Q2}e_\tau \quad (34b)$$

$$\begin{bmatrix} \dot{e}_{\omega_r} \\ \dot{e}_\tau \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & \frac{n^2}{J} \\ -k_{Q1} & -\frac{R_q}{L_f} - k_{Q2} \end{bmatrix}}_N \begin{bmatrix} e_{\omega_r} \\ e_\tau \end{bmatrix} \quad (34c)$$

$$tr(N) = -\frac{R_q}{L_f} - k_{Q2} \quad (34d)$$

$$det(N) = \frac{n^2}{J}k_{Q1} \quad (34e)$$

It is found that for all $k_{Q1} > 0$ and $k_{Q2} \geq 0$, the following conditions hold:

$$tr(N) < 0 \quad \text{and} \quad det(N) > 0,$$

indicating that N is a Hurwitz matrix.

The stability analysis confirms that for all selected control gains $k_{D1}, k_{D2}, k_{Q1}, k_{Q2}$ within their specified constraints, the system matrices remain Hurwitz. This ensures asymptotic stability of the controlled system, with error dynamics converging to zero. The results validate the effectiveness of the feedback linearization approach in achieving the desired performance.

3.2.3 Implementation and Tuning

The goal of this section is to determine the controller gains that ensure a fast and stable system response while respecting the motor's physical constraints. The approach consists of

analyzing the closed-loop characteristic equations and tuning the system parameters based on the natural frequency and damping ratio.

The state-space representation of the flux and direct current error dynamics is given by:

$$\begin{bmatrix} \dot{e}_\Phi \\ \dot{e}_{i_s^D} \end{bmatrix} = \begin{bmatrix} -\frac{1}{T_r} & \frac{R_{req}}{L_f} \\ \frac{1}{T_r L_f} - k_{D1} & -\frac{R_s + R_{req}}{L_f} - k_{D2} \end{bmatrix} \begin{bmatrix} e_\Phi \\ e_{i_s^D} \end{bmatrix} \quad (35)$$

The characteristic polynomial associated with this system is:

$$P(s) = s^2 + s \left(k_{D2} + \frac{1}{T_r} + \frac{R_s + R_{req}}{L_f} \right) + \frac{R_s}{T_r L_f} + k_{D1} - k_{D2} \quad (36a)$$

$$\omega_0 = \sqrt{\frac{R_s}{T_r L_f} + k_{D1} - k_{D2}} \quad (36b)$$

$$\zeta = \frac{\left(k_{D2} + \frac{1}{T_r} + \frac{R_s + R_{req}}{L_f} \right)}{2\sqrt{\frac{R_s}{T_r L_f} + k_{D1} - k_{D2}}} \quad (36c)$$

Similarly, for the speed dynamics:

$$\begin{bmatrix} \dot{e}_{\omega_r} \\ \dot{e}_\tau \end{bmatrix} = \begin{bmatrix} 0 & \frac{n^2}{J} \\ -k_{Q1} & -\frac{R_q}{L_f} - k_{Q2} \end{bmatrix} \begin{bmatrix} e_{\omega_r} \\ e_\tau \end{bmatrix} \quad (37)$$

$$P(s) = s^2 + s \left(k_{Q2} + \frac{R_q}{L_f} \right) + \frac{n}{J} k_{Q1} \quad (38a)$$

$$\omega_0 = \sqrt{\frac{n^2}{J} k_{Q1}} \quad (38b)$$

$$\zeta = \frac{\left(k_{Q2} + \frac{R_q}{L_f} \right)}{2\sqrt{\frac{n^2}{J} k_{Q1}}} \quad (38c)$$

The tuning process begins by selecting a suitable settling time. The objective is to minimize t_s while ensuring system stability and adhering to operational constraints, particularly not exceeding the maximum allowable voltage and current. Initially, the commonly used formula for underdamped systems was employed:

$$t_s = \frac{4}{\zeta \omega_0} \quad (39)$$

A settling time of $t_s = 0.01$ s was selected, representing a 75% reduction compared to the uncontrolled flux subsystem.

For the selection of the damping factor, the system's performance was observed to improve as the damping ratio was increased. This approach led to better stability, although excessive damping would slow the system down. Through empirical tuning, the damping ratios of $\zeta = 0.947$ for the flux subsystem and $\zeta = 5.72$ for the speed subsystem were chosen, which

provided an optimal balance between stability and response time without significantly delaying the system.

However, the overdamped behavior ($\zeta = 5.72$) renders the previous equation for underdamped systems inapplicable. Instead, the correct settling time equation for overdamped systems is:

$$t_s \approx \frac{4.6}{(\zeta - \sqrt{\zeta^2 - 1})\omega_0} \quad (40)$$

Although the initial settling time equation was based on an underdamped assumption, the revised calculation confirms that the selected gains yield a well-behaved response. The final tuning ensures both fast convergence and improved stability, demonstrating the effectiveness of the proposed controller while respecting operational constraints.

Finally, by solving for the controller gains based on the selected settling time and damping ratio, the following values were obtained:

$$k_{D1} = 2273, \quad k_{D2} = 618.444, \quad k_{Q1} = 2273, \quad k_{Q2} = 618.444.$$

3.2.4 Simulation results

The simulation is divided into three time intervals to analyze the system's response under different conditions:

- **Interval 1** ($t = 0$ to $t = 0.5$ s): The system is initialized at equilibrium, where all state variables are set to their nominal values.
- **Interval 2** ($t = 0.5$ to $t = 1.0$ s): A change in the reference speed is introduced to evaluate the controller's tracking performance.
- **Interval 3** ($t = 1.0$ to $t = 1.5$ s): At $t = 1.0$ s, an abrupt increase of 50% in the external torque is applied.

Where the system operates under the assumption that all motor states and external torque are perfectly known.

Initially, the system operates stably at its nominal equilibrium point, as anticipated. Upon introducing a change in the reference speed at $t = 0.5$ s, the system exhibits an overdamped response, with the state variables smoothly following the new reference (indicated by the red curve) and reaching a steady-state in approximately 0.02 s. When a sudden 50% increase in external torque is applied at $t = 1.0$ s, the system briefly deviates from its reference value, but the variation is minimal and barely perceptible. The system quickly returns to tracking the reference curve. This behavior demonstrates that, even with perfect knowledge of the motor states and external torque, the system takes a small but necessary adjustment period to stabilize. Throughout all conditions, no significant errors were observed, and the absolute errors remained within six decimal places, confirming the high precision and reliability of the control strategy.

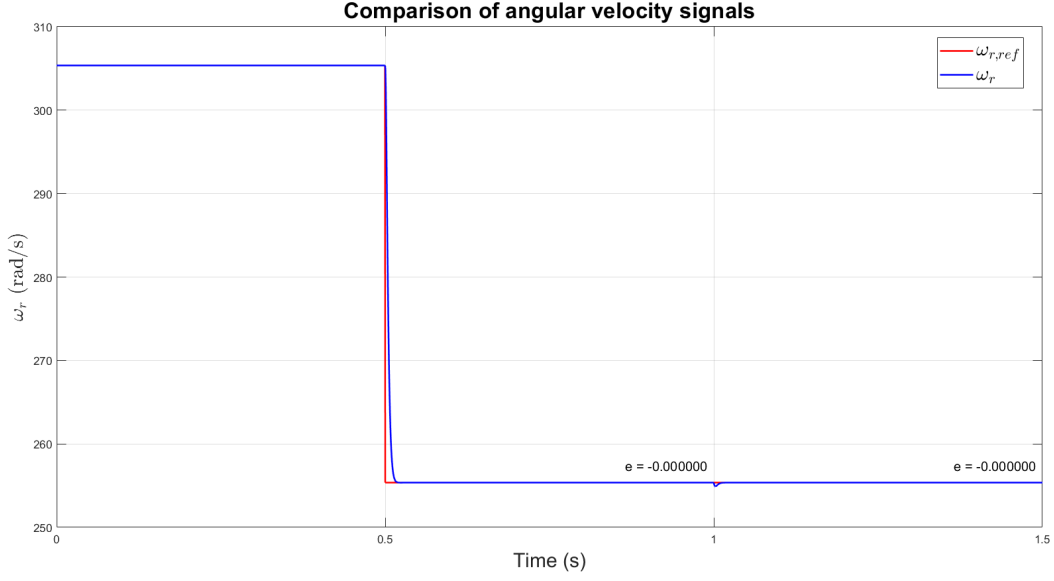


Figure 2: System response using feedback linearization-based control, assuming full state measurement and known external torque.

4 Observer

4.1 Motor equations

We begin by considering the induction motor system from Subsection 2.2 where we decompose into a cascade of two time-varying linear subsystems that independently govern the flux subsystem and the velocity subsystem. These subsystems can be rewritten from the induction motor equations (2) to represent with output injection terms two independent subsystems for the observer design:

- **Flow subsystem:** From knowledge of the input signals i_s^{xy} , u_s^{xy} and ω_e .
- **Velocity subsystem:** From the knowledge of the input signals i_s^{xy} , u_s^{xy} , and the ψ_s^{xy} signal obtained from the flow subsystem.

From here, the terms J_s and τ for the flow and velocity subsystems, respectively, will be obtained from the orthogonal projections of i_s^{xy} . In order to choose $(\psi_s^{xy}, \omega_r$ and $T_l)$ as state variables.

The dynamics of the flow subsystem are described by:

$$\frac{d\psi_s^{xy}}{dt} = u_s^{xy} - R_s i_s^{xy} - \omega_e \mathcal{J} \psi_s^{xy} \quad (41a)$$

$$\frac{d\phi}{dt} = 2 \langle \psi_s^{xy}, u_s^{xy} - R_s i_s^{xy} \rangle \quad (41b)$$

$$L_f \frac{dJ_s}{dt} = \frac{R_r}{L_r} \phi - R_+ J_s + \langle \psi_s^{xy}, u_s^{xy} \rangle - \frac{L_f}{2} R_- |i_s^{xy}|^2 \quad (41c)$$

$$0 = J_s - \langle i_s^{xy}, \psi_s^{xy} - \frac{L_f}{2} i_s^{xy} \rangle \quad (41d)$$

The dynamics of the velocity subsystem are given by:

$$L_f \frac{d\tau}{dt} = -R_t \tau - \langle \psi_s^{xy} - \sigma L_s i_s^{xy}, \omega_r \psi_s^{xy} + \mathcal{J} u_s^{xy} \rangle \quad (42a)$$

$$\frac{J}{n} \frac{dw_r}{dt} = n\tau - T_l \quad (42b)$$

$$\frac{dT_l}{dt} = 0 \quad (42c)$$

$$0 = \tau - \langle i_s^{xy}, \mathcal{J} \psi_s^{xy} \rangle \quad (42d)$$

where the linear variables in these equations were obtained in detail from [5]. As for the constants in the system, they are defined in the reference and do not require further explanation here.

4.2 Cascade Kalman filters

In this section, we present the two subsystems in flux and speed following a linear time-varying (LTV) system which is defined by time-dependent matrices $A(t)$, $B(t)$, $C(t)$, and $D(t)$.

The state-space representation of the LTV system is given by:

$$\dot{x}(t) = A(t)x(t) + b(t), \quad (43a)$$

$$0 = C(t)x(t) + d(t), \quad (43b)$$

Where the matrices $A(t) \in \mathbb{R}^{n \times n}$, $B(t) \in \mathbb{R}^{n \times 1}$, $C(t) \in \mathbb{R}^{p \times n}$, and $D(t) \in \mathbb{R}^{p \times 1}$.

The dynamics of the continuous-time Kalman filter for (43) follow:

$$\dot{\hat{x}}(t) = A(t)\hat{x}(t) + b(t) - L(t)(C(t)\hat{x}(t) + d(t)) \quad (44a)$$

$$L(t) = P(t)C(t)^T R^{-1} \quad (44b)$$

$$\dot{P}(t) = A(t)P(t) + P(t)A(t)^T + Q - P(t)C(t)^T R^{-1} C(t)P(t) \quad (44c)$$

where L is the filter gain matrix, belonging to the real numbers $\mathbb{R}^{n \times p}$, and $P(t)$ is the covariance matrix of the system, with $P(t) > 0$. Additionally, $Q \in \mathbb{R}^{n \times n}$ is the process noise covariance matrix ($Q \geq 0$), and $R \in \mathbb{R}^{p \times p}$ is the observation noise covariance matrix ($R > 0$), and D considered process noise weighting matrix ($D(t) = I$). These matrices model the noise in the process and observation, respectively.

To design the observer, we construct a cascaded structure from equations (44) where the first observer is dedicated to estimating the flux state $x_f = (\psi_s^{xy}, \phi, J_s)^T$, ensuring that $\hat{x}_f$ tends to x_f . The second observer then takes this estimated flux as an input to refine the estimation of the velocity state $x_v = (\tau, \omega_r, T_l)^T$, aiming for $\hat{x}_v$ to converge to x_v .

Observability is a critical property in control and estimation, determining whether the internal state of a system can be reconstructed from its outputs. For Linear Time-Varying (LTV) systems, where system matrices depend on time, traditional observability conditions are often inadequate, necessitating stronger frameworks such as differential observability and Uniform Complete Observability (UCO). These concepts ensure that the state can be uniquely and stably determined from the output and its derivatives, even as the system evolves over time. In this section, we formalize these ideas and establish their relationships, providing the foundation for robust state estimation and control in LTV systems.

Definition 1. Consider a LTV system

$$\dot{x} = A(t)x, \quad y = C(t)x,$$

where $x \in \mathbb{R}^n$ is the state, $y \in \mathbb{R}^m$ is the output, A is the system matrix, and C is the output matrix, where the observability matrix $\mathcal{O}_p(t)$, defined as

$$\mathcal{O}_p(t) = \begin{bmatrix} C(t) \\ \dot{C}(t) + C(t)A(t) \\ \ddot{C}(t) + \dot{C}(t)A(t) + C(t)A^2(t) \\ \vdots \\ C^{(p-1)}(t) + \sum_{i=0}^{p-2} C^{(i)}(t)A^{(p-1-i)}(t) \end{bmatrix},$$

is full rank (i.e., invertible) for all $t \geq 0$. This ensures that the state x can be uniquely determined from the outputs y and their derivatives up to order $p - 1$.

The system is Uniform Strongly Differentially Observable [8] of order p if there exists a positive constant ℓ such that:

$$\mathcal{O}_p(t)^\top \mathcal{O}_p(t) \geq \ell I > 0 \quad \forall t \geq 0.$$

The condition ensures that the observability mapping is injective, of Lipschitz type, and uniform in time.

Definition 2. The pair (A, C) in an LTV system is said to be Uniformly Completely Observable (UCO) if and only if there exist positive scalar constants δ_1 and γ_1 such that

$$\gamma_1^{-1} \leq \mathbf{d}^\top \mathcal{W}_0(t, t + \delta_1) \mathbf{d} \leq \gamma_1,$$

for all $t \geq t_0$ and $d \in \mathbb{R}^n$, with $\|d\| = 1$, where

$$\mathcal{W}_0(t_1, t_2) = \int_{t_1}^{t_2} \Phi^\top(\sigma, t_1) C^\top(\sigma) R^{-1}(\sigma) C(\sigma) \Phi(\sigma, t_1) d\sigma,$$

Here, $\Phi(\sigma, t_1)$ is the state transition matrix associated with $A(t)$, satisfying:

$$\frac{\partial}{\partial \sigma} \Phi(\sigma, t_1) = A(\sigma) \Phi(\sigma, t_1), \quad \Phi(t_1, t_1) = I,$$

where I is the identity matrix.[9]

The UCO property ensures that the observability Gramian is uniformly bounded and invertible over time, guaranteeing consistent state reconstruction. This property is closely related to the structure of the observability matrix $Q_o(t)$, which we define next to analyze observability at specific time instances.

Definition 3. The observability matrix [7] $Q_o(t)$ is defined below, where n is the dimension of x :

$$\begin{cases} Q_o(t) = [Q_0(t) & Q_1(t) & \dots & Q_{n-1}(t)] \\ Q_0(t) = C^T(t) \\ Q_{i+1}(t) = \dot{Q}_i(t) + A^T(t)Q_i(t) \end{cases}$$

Theorem 1. From [7] the system

$$\dot{x} = f(x, t) \quad , \quad y = h(x, t)$$

is UCO if there exist constants $\mu > 0$ and $m \in \mathbb{N}$ such that for all $t \geq 0$, the observability Gramian

$$\mathcal{O}(t) = (Q_0(t) \dots Q_m(t)) \begin{pmatrix} Q_0^T(t) \\ \vdots \\ Q_m^T(t) \end{pmatrix} \geq \mu I > 0.$$

Here, $Q_o(t)$ corresponds to $\mathcal{O}_p(t)$ in the defined LTV system.

From Theorem 1, we can establish a connection between Uniformly Strong (US) Differential Observability and Uniform Complete Observability (UCO). Specifically, the uniform lower bound on the observability Gramian ensures that the system satisfies the conditions for UCO,

4.2.1 Observability of flux subsystem

To demonstrate the convergence of $\hat{x}_f$ to x_f , we first need to define the observability of the flow subsystem. This step is necessary before we can proceed with proving the convergence. We will use the dq -reference frame due to the complexity involved in the differential observability demonstration in the α - β reference frame. Additionally, as we already know, u_s , i_s , and ω_s are known, and we take the output (41d) at order p .

$$J_s = \left\langle i_s^{dq}, \psi_s^{dq} - \frac{\sigma L_s}{2} i_s^{dq} \right\rangle \quad (45a)$$

$$\frac{R_r}{L_r} \|\psi_s^{dq}\|^2 + \langle \psi_s^{dq}, v_1^{dq} \rangle = v_0^{dq} \quad (45b)$$

$$\langle \psi_s^{dq}, v_2^{dq} \rangle - \omega_s \langle \mathcal{J} \psi_s^{dq}, v_1^{dq} \rangle = v_3^{dq} \quad (45c)$$

$$-\omega_s^2 \langle \psi_s^{dq}, v_1^{dq} \rangle - \omega_s \langle \mathcal{J} \psi_s^{dq}, v_2^{dq} \rangle = v_4^{dq} \quad (45d)$$

Based on the equations (1) expressed in a general frame, in the specific case when the reference frame velocity ω_e is equal to the stator speed ω_s , the model reduces to the dq -reference frame, the known variables $v_0 = f_1(i_s, u_s, \frac{di_s}{dt})$ and $v_1 = f_2(i_s, u_s, \frac{di_s}{dt})$ contain the derivative of the current, and successive derivatives related to v_3, v_4, v_5, v_6 and others involve infinitesimal derivatives of i_s and u_s . In this context, we can conclude that the worst-case scenario occurs when the motor operates in steady state, where all higher-order derivatives of these variables are zero. This leads to the observability determinant constraint being zero, representing the least favorable case. Therefore, we will apply the steady-state regime to analyze the most unfavorable case for the observability differential. Starting from the motor equations in the dq -frame, where i_s^{dq} , u_s^{dq} , and ω_s are constants, we obtain:

$$\omega_{s,\text{eq}} = \omega_{r,\text{ref}} + \frac{R_r T_{l,\text{eq}}}{n \|\psi_{r,\text{eq}}^{dq}\|^2} \quad (46a)$$

$$i_{r,\text{eq}}^{dq} = \frac{1}{R_r} (\omega_{r,\text{ref}} - \omega_{s,\text{eq}}) \mathcal{J} \psi_{r,\text{eq}}^{dq} \quad (46b)$$

$$i_{s,\text{eq}}^{dq} = \frac{1}{L_m} (\psi_{r,\text{eq}}^{dq} - L_r i_{r,\text{eq}}^{dq}) \quad (46c)$$

$$\psi_{s,\text{eq}}^{dq} = L_s i_{s,\text{eq}}^{dq} - L_m i_{r,\text{eq}}^{dq} \quad (46d)$$

$$u_{s,\text{eq}}^{dq} = R_s i_{s,\text{eq}}^{dq} + \omega_{s,\text{eq}} \mathcal{J} \psi_{s,\text{eq}}^{dq} \quad (46e)$$

Now, the signals v_i are known, specifically, v_1 and v_2 are given by:

$$v_{1,\text{eq}}^{dq} = u_{s,\text{eq}}^{dq} - R_+ i_{s,\text{eq}}^{dq} - \sigma L_s \omega_s \mathcal{J} i_{s,\text{eq}}^{dq} \quad (47a)$$

$$v_{2,\text{eq}}^{dq} = \frac{2R_r}{L_r} (u_{s,\text{eq}}^{dq} - R_s i_{s,\text{eq}}^{dq}) \quad (47b)$$

From equation (46e), we can rewrite the signal v_2 as:

$$v_{2,\text{eq}}^{dq} = \frac{2R_r}{L_r} (\omega_{s,\text{eq}} \mathcal{J} \psi_{s,\text{eq}}^{dq}) \quad (48)$$

Therefore, ψ_s^{dq} is uniquely determined if the following determinant from (45c) and (45d) is non zero:

$$(-\omega_{s,\text{eq}}^2 v_{1,\text{eq}}^{dq} + \omega_{s,\text{eq}} \mathcal{J} v_{2,\text{eq}}^{dq})^T \mathcal{J} (v_{2,\text{eq}}^{dq} + \omega_{s,\text{eq}} \mathcal{J} v_{1,\text{eq}}^{dq}) = \omega_{s,\text{eq}} \|\omega_{s,\text{eq}} v_{1,\text{eq}}^{dq} - \mathcal{J} v_{2,\text{eq}}^{dq}\|^2 \quad (49)$$

According to Lemma 2, referenced in [8], and by substituting (48) in (49), the flow subsystem is uniformly strongly differentially observable of order $p = 4$ if and only if, there exists ℓ positive such that:

$$\omega_{s,\text{eq}}^2 \|v_{1,\text{eq}}^{dq} + \frac{2R_r}{L_r} \psi_{s,\text{eq}}^{dq}\|^2 > \ell \quad (50)$$

From the equations in (46), we obtain:

$$\omega_{s,\text{eq}}^2 \left\| \frac{L_m}{L_r} \left(\frac{R_r}{L_r} \psi_{r,\text{eq}}^{dq} + \omega_{r,\text{ref}} \mathcal{J} \psi_{r,\text{eq}}^{dq} \right) \right\|^2 > \ell \quad (51)$$

In the worst-case scenario of US differential observability, as previously mentioned based on the signals v_1 and v_2 , the variables ω_s^{dq} and ψ_r^{dq} must be bounded. However, in the case where they are zero, observability is lost. This is particularly relevant in practical motor operation, as the flux ψ is zero when the motor is not operating.

From **Theorem 1**, we conclude that the flow subsystem is Uniformly Completely Observable due to the convergence properties derived from the successive derivatives of the output.

4.2.2 Observability of velocity subsystem

Now, in velocity subsystem (42), to demonstrate the convergence of $\hat{x}_v$ to x_v , we start from the observability differential, from the knowledge of u_s^{xy} , i_s^{xy} , ψ_s^{xy} and τ with

$$0 = \tau - \langle i_s^{xy}, \mathcal{J} \psi_s^{xy} \rangle \quad (52a)$$

$$\xi \omega_r = p_0^{xy} \quad (52b)$$

$$\omega_r \frac{d\xi}{dt} - \frac{n}{J} \xi T_l = p_1^{xy} \quad (52c)$$

$$\frac{d^2 \xi}{dt^2} \omega_r - \frac{2n}{J} \frac{d\xi}{dt} T_l = p_2^{xy} \quad (52d)$$

Where the signals p_i are known and we define

$$\xi = \langle \psi_s^{xy} - \sigma L_s i_s^{xy}, \psi_s^{xy} \rangle \quad (53)$$

Continuing differentiating equations (52) can be defined in matrix form, where k th time derived from the signal $\xi\omega_r$ represents

$$\begin{bmatrix} p_0^{xy} \\ p_1^{xy} \\ p_2^{xy} \\ \vdots \\ p_k^{xy} \end{bmatrix} = \begin{bmatrix} \xi & 0 \\ \frac{d\xi}{dt} & -\frac{n}{J}\xi \\ \frac{d^2\xi}{dt^2} & -\frac{2n}{J}\frac{d\xi}{dt} \\ \vdots & \vdots \\ \xi^{(k)} & -\frac{kn}{J}\xi^{(k-1)} \end{bmatrix} \begin{bmatrix} \omega_r \\ T_l \end{bmatrix}$$

According to Lemma 3 [8] for the case of uniform strongly differentially observable of order $k = 1$ if ξ and $\frac{d\xi}{dt}$ are bounded and there exists $l > 0$ such that for all $t \geq 0$,

$$\xi(t)^2 > l \quad (54)$$

For this case we will take as reference the steady state to analyze uniform strongly differentially observable, considering it as the most unfavorable scenario, similarly to the flow subsystem analysis, since we want to know for which precise cases there will be singularities where our system cannot be observable, then, u_s^{dq} , i_s^{dq} , ψ_s^{dq} are constants where (53) can be rewritten from (46) as:

$$\left(\frac{L_s}{L_r} \|\psi_r^{dq}\|^2\right)^2 > l \quad (55)$$

Then, the system is Uniformly Strongly (US) Differentially Observable of order $k = 1$ if ψ_r^{dq} is bounded. However, observability is lost if ψ_r^{dq} is zero. This is consistent with the worst-case scenario of US differential observability. Based on these results and the convergence properties derived from the output and its derivatives, we conclude that the velocity subsystem is UCO, as established in **Theorem 1**, in the same way as the flow subsystem.

4.3 Uniform Complete Controllability

Since we have already demonstrated that the pair (A, C) of the flow and velocity subsystem are UCO, we now proceed to prove that (A, C) is also Uniformly Completely Controllable (UCC).

Definition 4. *The pair (A, D) in an LTV system is said to be Uniformly Completely Controllable (UCC) if and only if there exist positive scalar constants $\delta_2 > 0$ and $\gamma_2 > 0$ such that:*

$$\gamma_2^{-1} \leq \mathbf{d}^\top \mathcal{W}_C(t, t + \delta_2) \mathbf{d} \leq \gamma_2,$$

for all $t \geq t_0$ and $\mathbf{d} \in \mathbb{R}^n$, with $\|\mathbf{d}\| = 1$, where the controllability Gramian $\mathcal{W}_C(t_1, t_2)$ is defined as:

$$\mathcal{W}_C(t_1, t_2) = \int_{t_1}^{t_2} \Phi(t_1, \sigma) D(\sigma) Q(\sigma) D^\top(\sigma) \Phi^\top(t_1, \sigma) d\sigma.$$

Here:

- $\Phi(t_1, \sigma)$ is the state transition matrix associated with $A(t)$.
- $D(\sigma)$ and $Q(\sigma)$ are matrices used to model process noise, with $Q(\sigma)$ being positive definite.

The Kalman filter (KF) in equation ((44)) can be derived from the matrices A , B , C , and D of the flow subsystem that are time-varying, which are known from equation ((41)).

$$A_f(t) = \begin{pmatrix} -\omega_e \mathcal{J} & 0 & 0 \\ 2(u_s^{xy} - R_s i_s^{xy})^T & 0 & 0 \\ \frac{1}{L_f}(u_s^{xy})^T & \frac{R_r}{L_f L_r} & -\frac{R_+}{L_f} \end{pmatrix}, \quad b_f(t) = \begin{pmatrix} u_s^{xy} - R_s i_s^{xy} \\ 0 \\ -\frac{R_-}{2}|i_s^{xy}|^2 \end{pmatrix}.$$

$$C_f(t) = \begin{pmatrix} -(i_s^{xy})^T & 0 & 1 \end{pmatrix}, \quad d_f(t) = \frac{L_f}{2}|i_s^{xy}|^2.$$

Since the LTV system is bounded, there exist $(\alpha_1, \alpha_2) \in \mathbb{R}$ such that

$$\|\mathbf{A}_f(\mathbf{t})\| \leq \alpha_1, \quad \|\mathbf{Q}_f(\mathbf{t})\| = \alpha_2, \quad \forall t \in [0, m]. \quad (56)$$

It means that, for all (t, s) ,

$$\exp(-\alpha_1|t - s|) \leq \|\Phi_f(t, s)\| \leq \exp(\alpha_1|t - s|). \quad (57)$$

And the controllability Gramian of (A, D) is given by:

$$\mathcal{W}_{C_f}(t_1, t_2) = \int_{t_1}^{t_2} \Phi_f(t_1, \sigma) \mathbf{D}_f(\sigma) \mathbf{Q}_f(\sigma) \mathbf{D}_f^\top(\sigma) \Phi_f^\top(t_1, \sigma) d\sigma. \quad (58)$$

Where the simplified norm of the controllability Gramian is bounded by:

$$\|\mathcal{W}_{C_f}(t_1, t_2)\| \leq \int_{t_1}^{t_2} \|\Phi_f(t_1, \sigma)\| \|\mathbf{Q}_f(\sigma)\| \|\Phi_f^\top(t_1, \sigma)\| d\sigma. \quad (59)$$

As the controllability Gramian is a definite integral of a bounded system, it follows that the pair (A_f, D_f) is UCC, so there exists a positive value γ_2 such that:

$$\gamma_2^{-1} \leq \|\mathcal{W}_{C_f}(t_1, t_2)\| \leq \gamma_2. \quad (60)$$

For velocity subsystem:

$$A_v := \begin{pmatrix} -\frac{R_t}{L_f} & \langle \psi_s^{xy}, i_s^{xy} - \frac{1}{L_f} \psi_s^{xy} \rangle & 0 \\ \frac{n^2}{J} & 0 & -\frac{n}{J} \\ 0 & 0 & 0 \end{pmatrix}, \quad b_v := \begin{pmatrix} b_1 \\ 0 \\ 0 \end{pmatrix}$$

$$C_v := \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}, \quad d_v := -\langle i_s^{xy}, \widehat{\mathcal{J} \psi_s^{xy}} \rangle.$$

With the same LTV system for velocity bounded and ψ_s^{xy} known, there exist $(\alpha_3, \alpha_4) \in \mathbb{R}$ such that

$$\|\mathbf{A}_v(\mathbf{t})\| \leq \alpha_3, \quad \|\mathbf{Q}_v(\mathbf{t})\| = \alpha_4, \quad \forall t \in [0, m]. \quad (61)$$

And, for all (t, s) ,

$$\exp(-\alpha_3|t - s|) \leq \|\Phi_v(t, s)\| \leq \exp(\alpha_3|t - s|). \quad (62)$$

The result is exactly the same as for the velocity subsystem, so the pair (A_v, D_v) is UCC if there exist positive scalars (δ_4, γ_4) such that

$$\gamma_4^{-1} \leq \mathbf{d}^\top \mathcal{W}_C(t, t + \delta_4) \mathbf{d} \leq \gamma_4. \quad (63)$$

The results obtained for UCO, UCC in (A, C) and (A, D) respectively will allow us to demonstrate UCO and UCC in $(\hat{A}, \hat{C})$ and $(\hat{A}, \hat{D})$ respectively, from the following velocity observer.

$$\begin{cases} \dot{\hat{x}}_v(t) = \hat{A}_v(t)\hat{x}_v(t) + \hat{b}_v(t) - \hat{L}_v(t)(\hat{C}_v(t)\hat{x}_v(t) + \hat{d}_v(t)) \\ \dot{\hat{L}}_v(t) = \hat{P}_v(t)\hat{C}_v(t)^T R_v^{-1} \\ \dot{\hat{P}}_v(t) = \hat{A}_v(t)\hat{P}_v(t) + \hat{P}_v(t)\hat{A}_v(t)^T + D_v(t)Q_v D_v(t)^T - \hat{P}_v(t)\hat{C}_v(t)^T R_v^{-1} \hat{C}_v(t)\hat{P}_v(t) \end{cases}$$

From the Kalman filter implemented for the velocity subsystem where the errors decay exponentially fast with time, there exist positive scalar constants $\alpha_9, \alpha_{10}, \alpha_{11}, \alpha_{12}, \lambda_1, \lambda_2, \lambda_3$ and λ_4 such that for all $t \geq t_0$.

$$\begin{cases} \tilde{A}_v := A_v - \hat{A}_v, \\ \tilde{b}_v := b_v - \hat{b}_v, \\ \tilde{C}_v := C_v - \hat{C}_v, \\ \tilde{d}_v := d_v - \hat{d}_v, \\ \tilde{D}_v := D_v - \hat{D}_v, \\ \tilde{u}_v := u_v - \hat{u}_v, \\ \tilde{y}_v := y_v - \hat{y}_v. \end{cases} \quad \text{and} \quad \begin{cases} \|\tilde{\mathbf{A}}_v\| \leq \alpha_9 e^{-\lambda_1(t-t_0)} \\ \|\tilde{\mathbf{b}}_v\| \leq \alpha_{10} e^{-\lambda_2(t-t_0)} \\ \|\tilde{\mathbf{C}}_v\| = 0 \\ \|\tilde{\mathbf{d}}_v\| \leq \alpha_{11} e^{-\lambda_3(t-t_0)} \\ \|\tilde{\mathbf{D}}_v\| = 0 \\ \|\tilde{\mathbf{u}}_v\| = 0 \\ \|\tilde{\mathbf{y}}_v\| \leq \alpha_{12} e^{-\lambda_4(t-t_0)} \end{cases}$$

As the dynamic errors decay exponentially, we confirm from the article by Viegas [9] that the pairs $(\hat{A}, \hat{C})$ and $(\hat{A}, \hat{D})$ are UCO and UCC, respectively.

4.4 Exponential Stability

The stability analysis of the flow and velocity subsystems will now be performed, based on [6].

Theorem 2. *Assume that the model of the message process by LTV system is:*

- (i) *uniformly completely observable;*
- (ii) *uniformly completely controllable;*
- (iii) $a_3 \leq \|Q(t)\| \leq a_4$,
- (iv) $a_5 \leq \|R(t)\| \leq a_6$ for all t ;
- (v) $\|A(t)\| \leq a_7$.

with a_3, a_4, a_5, a_6 , and a_7 being positive constants.

Then the following is true:

- (i) *The optimal filter is uniformly asymptotically stable and implies exponential stability by the Perron-Liapunov-Malkin theorem.*

From Theorem 2, given that the conditions UCO(i) and UCC(ii) are satisfied in the flux subsystem, as well as the boundedness conditions (iii), (iv) and (v), we demonstrate that:

$$\|(\hat{\psi}_s, \hat{\phi}, \hat{j}_s) - (\psi_s, \phi, j_s)\| \leq \alpha e^{-\lambda(t-t_0)} \quad \forall t \geq t_0$$

where $\alpha > 0$ and $\lambda > 0$ are positive constants.

Since

$$|\hat{\psi}_s^{xy} - \psi_s^{xy}| \leq \alpha e^{-\lambda(t-t_0)},$$

and given that the conditions UCO(i) and UCC(ii) are satisfied in the flow subsystem, as well as the boundedness conditions (iii), (iv), and (v), we demonstrate that:

$$\|(\hat{\omega}_r, \hat{T}_l, \hat{\tau}) - (\omega_r, T_l, \tau)\| \leq \alpha' e^{-\lambda'(t-t_0)} \quad \forall t \geq t_0$$

where $\alpha' > 0$ and $\lambda' > 0$ are positive constants.

5 Output Feedback Control and Robustness

In this section, we focus on the interaction between the observer and the controller in a generalized nonlinear system. The system's dynamics are represented by a set of nonlinear equations, where the functions $f(x)$ and $g(x)$ depend on the state x . The goal is to design a controller that can drive the system to a desired state, despite the presence of estimation errors in both the system states and the exogenous parameters. By incorporating the observer dynamics into the control law, the system compensates for inaccuracies in the estimation of these parameters.

We analyze the error dynamics for this generalized nonlinear system, deriving a control law that considers the nonlinearities and ensures robust performance even when the estimates of the system's state and external parameters are not perfect. To establish the stability of the system under these conditions, we use Lyapunov's method, which provides a mathematical framework to guarantee convergence to the desired equilibrium point, even in the presence of estimation errors.

5.1 Observer and Controller Interaction

Consider the generalized system dynamics $\dot{x} = Ax + B(f(x) + g(x)u) + D(p)$, where $x \in \mathbb{R}^2$, and $f(x)$ and $g(x)$ are nonlinear state-dependent functions. The term p represents an exogenous parameter that is independent of the system states and does not evolve according to the system dynamics. Let $(x_{ref}(p), u_{ref}(p))$ be constants such that:

$$Ax_{ref}(p) + Bu_{ref}(p) + D(p) = 0,$$

and define the error as $e = x - x_{ref}(p)$.

- If x and p are known, then $u = \frac{1}{g(x)} (-f(x) + u_{ref}(p) - K(x - x_{ref}(p)))$.
- If only the estimates $\hat{x}$ and $\hat{p}$ of the state x and the parameter p , respectively, are available then $u = \frac{1}{g(\hat{x})} (-f(\hat{x}) + u_{ref}(\hat{p}) - K(\hat{x} - x_{ref}(\hat{p})))$.
with $\Delta x = x - \hat{x}$ and with $\Delta p = p - \hat{p}$.
Consequently, the error dynamics is given by:

$$\dot{e} = Ax + B \left(f(x) - f(\hat{x}) \frac{g(x)}{g(\hat{x})} + u_{ref}(\hat{p}) \frac{g(x)}{g(\hat{x})} - K(e + x_{ref}(p) - x_{ref}(\hat{p}) - \Delta x) \frac{g(x)}{g(\hat{x})} \right) + D(p) \quad (64a)$$

$$\begin{aligned} \dot{e} = & Ax + B \left(f(x) - f(\hat{x}) \frac{g(x)}{g(\hat{x})} + u_{ref}(\hat{p}) \frac{g(x)}{g(\hat{x})} - u_{ref}(p) - K(x_{ref}(p) - x_{ref}(\hat{p}) - \Delta x) \frac{g(x)}{g(\hat{x})} \right. \\ & \left. + u_{ref}(p) - Ke - Ke \left(\frac{g(x)}{g(\hat{x})} - 1 \right) \right) + D(p) \end{aligned} \quad (64b)$$

$$\dot{e} = A(x_{ref}(p) + e) + Bu_{ref}(p) - BKe + B\Theta(x, \hat{x}, p, \hat{p}) + D(p) \quad (64c)$$

$$\dot{e} = Ax_{ref}(p) + Bu_{ref}(p) + D(p) + Ae - BKe + B\Theta(x, \hat{x}, p, \hat{p}) \quad (64d)$$

$$\dot{e} = (A - BK)e + B\Theta(x, \hat{x}, p, \hat{p}) \quad (64e)$$

Where the control gain matrix K is designed such that $(A - BK)$ is Hurwitz, and the function $\Theta(x, \hat{x}, p, \hat{p})$, introduced in the error dynamics, is defined by:

$$\Theta(x, \hat{x}, p, \hat{p}) = f(x) - f(\hat{x}) \frac{g(x)}{g(\hat{x})} + u_{ref}(\hat{p}) \frac{g(x)}{g(\hat{x})} - u_{ref}(p) - K(x_{ref}(p) - x_{ref}(\hat{p}) - \Delta x) \frac{g(x)}{g(\hat{x})} - Ke \left(\frac{g(x)}{g(\hat{x})} - 1 \right),$$

Furthermore, recalling that $\Delta x = x - \hat{x}$ and $\Delta p = p - \hat{p}$, the function $\Theta(x, \hat{x}, p, \hat{p})$ satisfies the following properties:

$$\lim_{(\Delta x, \Delta p) \rightarrow (0,0)} \Theta(x, \hat{x}, p, \hat{p}) = 0 \quad \text{and} \quad \Theta(x, \hat{x}, p, \hat{p}) = 0 \quad \text{when} \quad \Delta x = 0 \wedge \Delta p = 0.$$

Additionally, it is assumed that $|\Theta(x, \hat{x}, p, \hat{p})| \leq L(|\Delta x| + |\Delta p|)$, where L is a positive constant. This implies that $\Theta(x, \hat{x}, p, \hat{p})$ vanishes as the estimation errors Δx and Δp diminish.

5.2 Stability Validation Using Lyapunov's Method

The error dynamics

$$\dot{e} = (A - BK)e + B\Theta(x, \hat{x}, p, \hat{p})$$

indicates that the system is influenced by an external disturbance-like term Θ . However, this expression alone does not directly ensure stability at the origin. To establish stability, we employ a Lyapunov function approach. Consider the following Lyapunov candidate function:

$$V(e) = e^T P e. \quad (65)$$

with the following time derivative:

$$\dot{V} = \dot{e}^T P e + e^T P \dot{e}, \quad (66)$$

where P is a symmetric positive definite matrix. Given that $A - BK$ is designed to be Hurwitz, there exist $P > 0$ and $a > 0$ such that:

$$(A - BK)^T P + P(A - BK) \leq -aP. \quad (67)$$

Therefore, the time derivative of $V(e)$ can be bounded as follows:

$$\dot{V} \leq -aV + 2e^T P \Theta. \quad (68)$$

Applying Young's inequality:

$$x^T P y \leq \frac{\alpha}{2} x^T P x + \frac{1}{2\alpha} y^T P y, \quad (69)$$

and choosing $\alpha = \frac{a}{2}$, we obtain:

$$\dot{V} \leq -aV + \frac{a}{2} \underbrace{e^T P e}_V + \frac{2}{a} B^T \Theta P B \Theta. \quad (70)$$

Furthermore, using the spectral properties of positive definite matrices, we apply the inequality:

$$B^T \Theta P B \Theta \leq \lambda_{\max}(P) |B \Theta|^2. \quad (71)$$

Thus, the Lyapunov function derivative satisfies:

$$\dot{V} \leq -\frac{a}{2} V + \frac{2}{a} \lambda_{\max}(P) |B \Theta|^2. \quad (72)$$

$$\dot{V} \leq -\frac{a}{2} V + \frac{2}{a} \lambda_{\max}(P) |B|^2 |\Theta|^2. \quad (73)$$

$$\dot{V} \leq -\frac{a}{2} V + \gamma(|\Theta|)$$

where

$$\gamma(|\Theta|) = \frac{2}{a} \lambda_{\max}(P) |B|^2 |\Theta|^2$$

is a class $\mathcal{K}$ function.

This satisfies property 5 of Theorem 3.38 from Praly and Bresh-Pietri [1], thereby proving that the system exhibits Input-to-State Stability (ISS) with respect to the perturbation Θ . This implies that the system state e remains bounded for bounded disturbances and vanishes asymptotically when $\Theta \rightarrow 0$.

Now, recalling the assumption

$$|\Theta| \leq L(|\Delta x| + |\Delta p|),$$

we observe that the perturbation term Θ remains bounded by the estimation errors Δx and Δp . Consequently, for Θ to vanish asymptotically, it is necessary that both estimation errors Δx and Δp converge to zero.

This implies that the system's robustness and stability are directly linked to the accuracy of state and parameter estimation. In other words, achieving asymptotic stability requires that the estimation errors diminish over time. If the estimation process ensures that $\Delta x \rightarrow 0$ and $\Delta p \rightarrow 0$, then the perturbation Θ also vanishes, reinforcing the system's stability. Therefore, under these conditions, the system exhibits Input-to-State Stability (ISS) with respect to estimation errors, ensuring that any initial deviation due to uncertainties or perturbations eventually decays, leading to a stable trajectory.

The necessary conditions for applying the stability analysis were previously established in the observer analysis section. Specifically, the observability conditions and the convergence properties of the flux and velocity subsystems were rigorously defined, providing the foundation for the subsequent stability analysis. With these conditions in place, we were able to proceed

with the current analysis, which demonstrates the system's robustness and stability in the presence of estimation errors.

Moreover, the control gain matrix K plays a crucial role in determining both the rate of convergence and the stability margin of the system. Since the parameter a in the Lyapunov derivative is related to the eigenvalues of $A - BK$, an appropriate choice of K can enhance stability by increasing a , leading to a stronger negative term in the Lyapunov function derivative. This results in faster error convergence and improved robustness against perturbations.

In summary, the error converges to the origin as the estimation errors Δx and Δp vanish, a characteristic of the system demonstrated in section 4, which makes the presented analysis feasible. Additionally, selecting K to maximize a strengthens stability, ensuring faster stabilization and better system performance.

Finally, this analysis not only establishes the general convergence of the error under the proposed control law but also confirms its applicability to the two real systems under study. This result guarantees that errors with respect to the reference trajectory diminish over time, even in the presence of observer estimation errors.

5.3 Application to singular perturbations control

In the context of singular perturbations, it is not immediately obvious that the system under study adheres to the general form previously described. This is primarily due to the presence of estimation errors in the currents i_s^D and i_s^Q , which arise from the transformation based on the estimated flux ($\hat{\Phi}$). However, by carefully analyzing the error dynamics, we can derive an expression that aligns with the general structure, thereby validating the subsequent stability analysis.

Building upon the system equations presented in 7 and substituting the control signal from 8 and 9, while accounting for estimation errors—including those in the currents i_s^D and i_s^Q , which are derived through a transformation based on the estimated Φ —the resulting system is obtained as follows:

- For i_s^D and i_s^Q :

$$L_f \frac{di_s^D}{dt} = -(R_s + R_{req}) i_s^D + \frac{1}{T_r} \Phi + L_f R_{req} \frac{i_s^{Q^2}}{\Phi} + L_f \omega_r i_s^Q - K_P^D \left(i_s^{\hat{D}} - i_{s,ref}^{\hat{D}} \right) - \eta_D \quad (74a)$$

$$\frac{d\eta_D}{dt} = K_I^D \left(i_s^{\hat{D}} - i_{s,ref}^{\hat{D}} \right) \quad (74b)$$

$$L_f \frac{di_s^Q}{dt} = -(R_s + R_{req}) i_s^Q - \omega_r \Phi - L_f \omega_r i_s^D - L_f R_{req} \frac{i_s^D i_s^Q}{\Phi} - K_P^Q \left(i_s^{\hat{Q}} - i_{s,ref}^{\hat{Q}} \right) - \eta_Q \quad (74c)$$

$$\frac{d\eta_Q}{dt} = K_I^Q \left(i_s^{\hat{Q}} - i_{s,ref}^{\hat{Q}} \right) \quad (74d)$$

We define $\Delta i_s^D = i_s^D - i_s^{\hat{D}}$ and $\Delta i_s^Q = i_s^Q - i_s^{\hat{Q}}$. And according to subsection 3.1.2, we can conclude that i_s^D and i_s^Q rapidly converge to $(\Delta i_s^D + i_{s,ref}^{\hat{D}})$ and $(\Delta i_s^Q + i_{s,ref}^{\hat{Q}})$, respectively.

$$L_f \frac{d\iota_s^D}{dt} = F(x) - K_P^D \left(\iota_s^D - (\Delta\iota_s^D + \hat{\iota}_{s,\text{ref}}^D) \right) - \eta_D \quad (75a)$$

$$\frac{d\eta_D}{dt} = K_I^D \left(\iota_s^D - (\Delta\iota_s^D + \hat{\iota}_{s,\text{ref}}^D) \right) \quad (75b)$$

$$L_f \frac{d\iota_s^Q}{dt} = G(x) - K_P^Q \left(\hat{\iota}_s^Q - (\Delta\iota_s^Q + \hat{\iota}_{s,\text{ref}}^Q) \right) - \eta_Q \quad (75c)$$

$$\frac{d\eta_Q}{dt} = K_I^Q \left(\hat{\iota}_s^Q - (\Delta\iota_s^Q + \hat{\iota}_{s,\text{ref}}^Q) \right) \quad (75d)$$

This allows us to substitute them into the speed control loop expressions:

- For Φ :

$$\frac{d\Phi}{dt} = -\frac{1}{T_r} \Phi + R_{req}(\Delta\iota_s^D + \hat{\iota}_{s,\text{ref}}^D) \quad (76a)$$

$$\frac{d\Phi}{dt} = -\frac{1}{T_r} \Phi + R_{req} \left(\frac{1}{R_{req}} \right) \left(\frac{1}{T_r} \Phi_{ref} - \lambda_1 (\hat{\Phi} - \Phi_{ref}) \right) + R_{req} \Delta\iota_s^D \quad (76b)$$

$$\frac{d\Phi}{dt} = -\frac{1}{T_r} \Phi + \frac{1}{T_r} \Phi_{ref} - \lambda_1 (\Phi - \Delta\Phi - \Phi_{ref}) + R_{req} \Delta\iota_s^D \quad (76c)$$

$$\frac{d\Phi}{dt} = -\left(\frac{1}{T_r} + \lambda_1 \right) (\Phi - \Phi_{ref}) - \lambda_1 \Delta\Phi + R_{req} \Delta\iota_s^D \quad (76d)$$

$$\frac{de_\Phi}{dt} = -\left(\frac{1}{T_r} + \lambda_1 \right) e_\Phi - \lambda_1 \Delta\Phi + R_{req} \Delta\iota_s^D \quad (76e)$$

- For ω_r :

$$\frac{J}{n} \frac{d\omega_r}{dt} = n\Phi(\Delta\iota_s^Q + \hat{\iota}_{s,\text{ref}}^Q) - T_l \quad (77a)$$

$$\frac{J}{n} \frac{d\omega_r}{dt} = n\Phi \frac{1}{n\hat{\Phi}} \left(\hat{T}_l - \lambda_2 (\hat{\omega}_r - \omega_{r,\text{ref}}) \right) - T_l + n\Phi \Delta\iota_s^Q \quad (77b)$$

$$\frac{J}{n} \frac{d\omega_r}{dt} = \frac{\Phi}{\hat{\Phi}} (T_l - \Delta T_l - \lambda_2 (\omega_r - \Delta\omega_r - \omega_{r,\text{ref}})) - T_l + n\Phi \Delta\iota_s^Q \quad (77c)$$

$$\frac{J}{n} \frac{de_{\omega_r}}{dt} = \frac{\Phi}{\hat{\Phi}} (T_l - \Delta T_l - \lambda_2 (e_{\omega_r} - \Delta\omega_r)) - T_l + n\Phi \Delta\iota_s^Q \quad (77d)$$

$$\frac{J}{n} \frac{de_{\omega_r}}{dt} = -\lambda_2 e_{\omega_r} + (T_l - \lambda_2 e_{\omega_r}) \left(\frac{\Phi}{\hat{\Phi}} - 1 \right) - \frac{\Phi}{\hat{\Phi}} (\Delta T_l - \lambda_2 \Delta\omega_r) + n\Phi \Delta\iota_s^Q \quad (77e)$$

From these expressions, it becomes evident that the error dynamics for both the flux and the rotor speed share a common structure. Specifically, each error equation consists of two main components:

An exponentially stable term, which ensures that the error decays exponentially in the absence of perturbations, and a perturbation term that vanishes as the estimation errors tend to zero and is identically zero when the estimation errors are null.

This structure aligns with the general form of singularly perturbed systems. Consequently, the stability analysis presented earlier remains valid, and the system exhibits Input-to-State Stability (ISS) with respect to the estimation errors.

5.3.1 Sensitivity results to perturbations

The simulation is divided into three time intervals to analyze the system's response under different conditions:

- **Interval 1** ($t = 0$ to $t = 0.5$ s): The system is initialized at equilibrium, where all state variables are set to their nominal values.
- **Interval 2** ($t = 0.5$ to $t = 1.0$ s): A change in the reference speed is introduced to evaluate the controller's tracking performance.
- **Interval 3** ($t = 1.0$ to $t = 1.5$ s): An abrupt increase of 50% in the external torque is applied.

Here, the singular perturbation-based controller uses the estimated values of the motor states and the external torque. The following graph illustrates the system's response under the different time intervals described earlier. The red curve represents the reference speed, the blue curve shows the actual motor speed, and the green curve corresponds to the estimated speed.

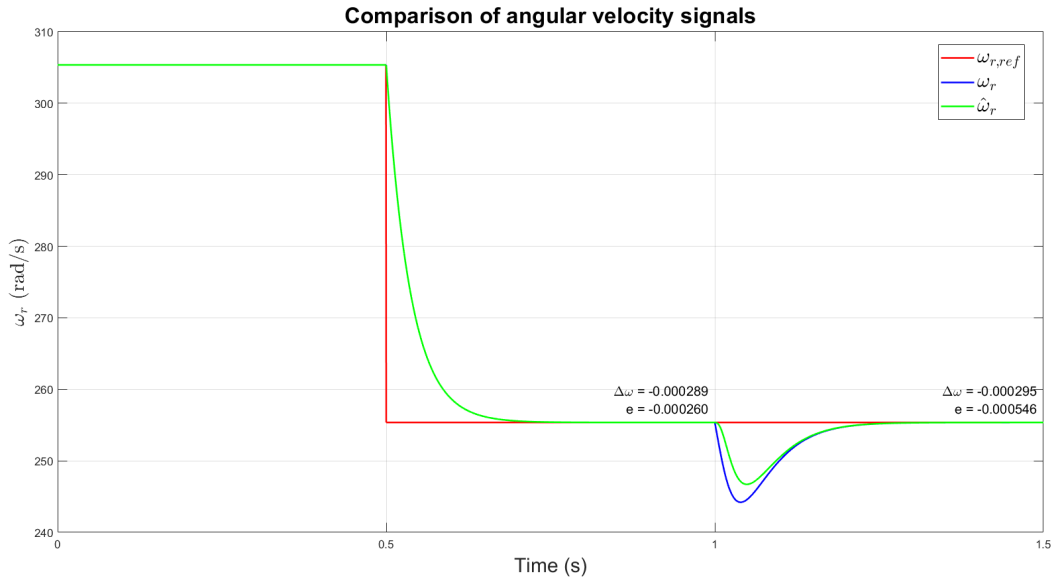


Figure 3: Performance of the output feedback controller using the observer.

The results demonstrate that when the system and observer are initialized at equilibrium, stability is maintained as expected. In the second interval, the system follows the reference trajectory with an overdamped response, reaching the desired speed in approximately 0.25 s. The observer, given appropriate initial conditions, accurately estimates the real system states without deviation.

A notable difference arises in the third interval, where a significant disturbance is observed following the variation in external torque. Unlike the ideal case, the observer requires approximately 0.25 s to readjust and correctly estimate the system states. Subsequently, within another 0.25 s, the system successfully returns to the reference speed.

Regarding absolute errors, $\Delta\omega$ represents the estimation error, while e denotes the deviation from the reference. These errors are in the order of 10^{-4} , suggesting that they likely result from numerical inaccuracies within the simulation software, as they decrease when refining computational tolerances.

5.3.2 Sensitivity results to modeling errors

The simulation conditions and time intervals remain the same as those presented in 5.3.1. In this section, we analyze the system's response under model uncertainties by considering two cases: one with a 10% error and another with a 25% error. These errors are directly applied to the four parameters that define the system, as specified in 6.

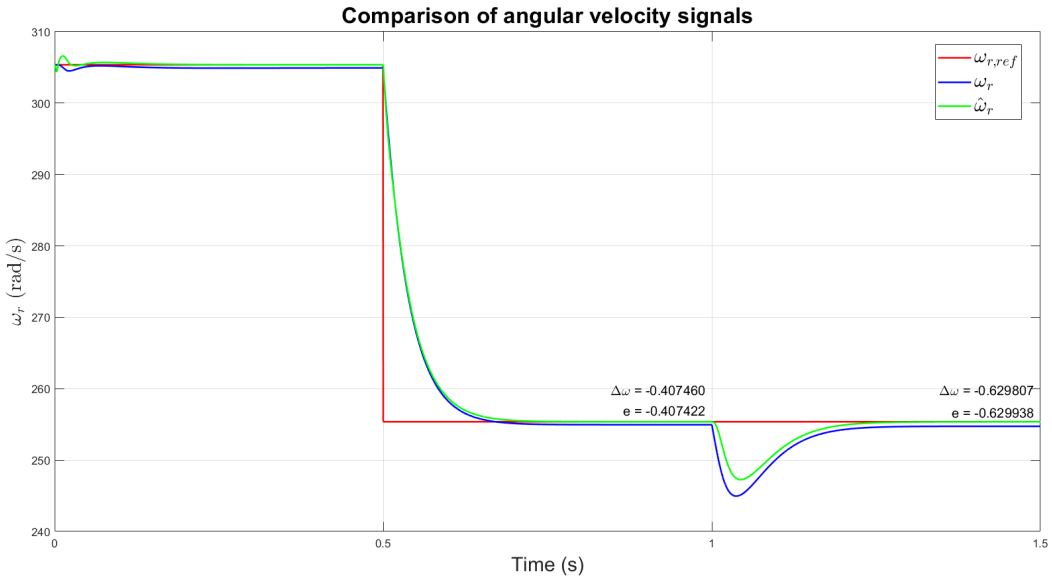


Figure 4: Controller response with a 10% modeling error in the motor parameters

At the beginning of the simulations, the systems deviates from their initial equilibrium, and the estimated value noticeably diverges from the actual value. This divergence is particularly pronounced in the second case. After approximately 0.1 s in the first case and 0.2 s in the second case, the estimated system reaches equilibrium and stabilizes. However, a steady-state error with respect to the actual motor speed is observed, which increases as the modeling error becomes larger.

In the second and third intervals, the system exhibits similar behavior to the simulation without modeling errors, with comparable settling times. However, a key discrepancy is that the estimated value attempts to track the actual value but consistently retains an error. Once stabilization occurs, a steady-state estimation error persists, preventing the system from fully reaching the reference speed. The estimation error is approximately 0.4 rad/s in the first case and 0.5 rad/s in the second case. These results indicate that the robustness of the controller is affected by the accuracy of the model. Nevertheless, even with a 25% parameter error, the observed deviations remain below 0.2% of the target speed, demonstrating that the controller maintains satisfactory performance under significant modeling uncertainties.

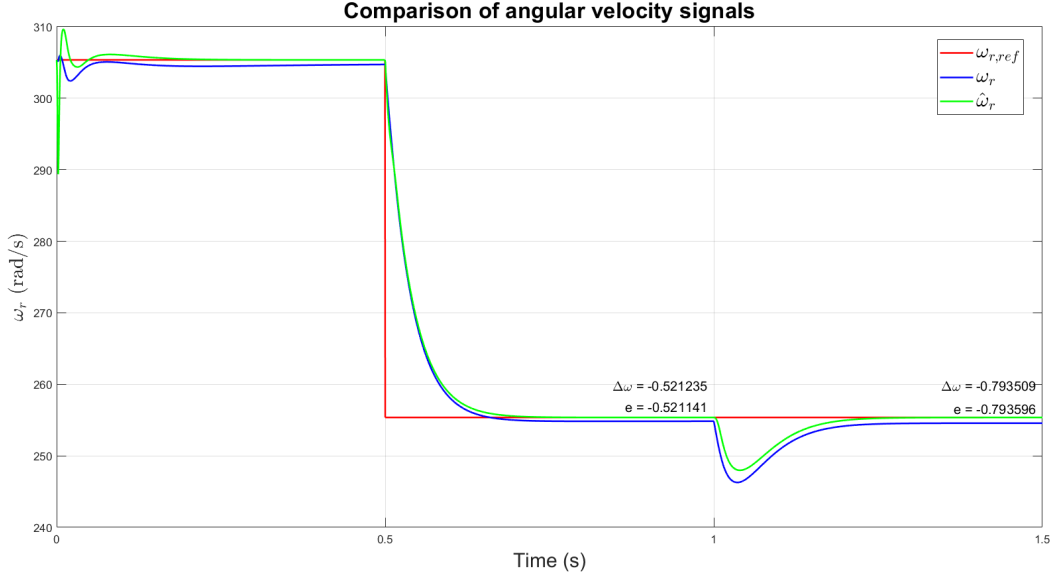


Figure 5: Controller response with a 25% modeling error in the motor parameters

5.4 Application to feedback linearization control

The feedback linearization approach yields a system structure that directly aligns with the general case described earlier. From the equations in 15, we obtain the following system:

$$\begin{bmatrix} \dot{\Phi} \\ \dot{i}_s^D \end{bmatrix} = \begin{bmatrix} -\frac{1}{T_r} & -\frac{R_{req}}{L_f} \\ \frac{1}{T_r L_f} & -\frac{R_s + R_{req}}{L_f} \end{bmatrix} \begin{bmatrix} \Phi \\ i_s^D \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \left(L_f R_{req} \frac{i_s^{Q^2}}{\Phi} + L_f \omega_r i_s^Q + u_s^D \right), \quad (78)$$

This system describes the dynamics of two states, Φ and i_s^D , with the control input embedded in the nonlinear term. The structure of this system closely resembles the general form previously analyzed, where the state dynamics are governed by a linear state matrix and a nonlinear control input term.

Similarly, from the equations in 22, we derive the following system for ω_r and τ :

$$\begin{bmatrix} \dot{\omega}_r \\ \dot{\tau} \end{bmatrix} = \begin{bmatrix} 0 & \frac{n^2}{J} \\ 0 & -\frac{R_q}{L_f} \end{bmatrix} \begin{bmatrix} \omega_r \\ \tau \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} (\Phi(\Phi + L_f i_s^D) \omega_r + \Phi u_s^Q) + \begin{bmatrix} -\frac{n}{J} T_l \\ 0 \end{bmatrix}. \quad (79)$$

Here, the system dynamics are similarly characterized by a linear state matrix and a nonlinear control input term, along with an additional term accounting for the load torque T_l . This structure is consistent with the general formulation used in the earlier Lyapunov-based stability analysis.

In both cases, the nonlinear control terms appear as additive components in the input vector, reflecting the feedback linearization approach. This alignment with the general case confirms that the stability analysis techniques previously developed—particularly the Lyapunov-based approach—are directly applicable to these systems.

5.4.1 Sensitivity results to perturbations

The simulation is divided into three time intervals to analyze the system's response under different conditions:

- **Interval 1** ($t = 0$ to $t = 0.5$ s): The system is initialized at equilibrium, where all state variables are set to their nominal values.
- **Interval 2** ($t = 0.5$ to $t = 1.0$ s): A change in the reference speed is introduced to evaluate the controller's tracking performance.
- **Interval 3** ($t = 1.0$ to $t = 1.5$ s): An abrupt increase of 50% in the external torque is applied.

Here, the feedback linearization-based controller uses the estimated values of the motor states and the external torque. The following graph illustrates the system's response under the different time intervals described earlier. The red curve represents the reference speed, the blue curve shows the actual motor speed, and the green curve corresponds to the estimated speed.

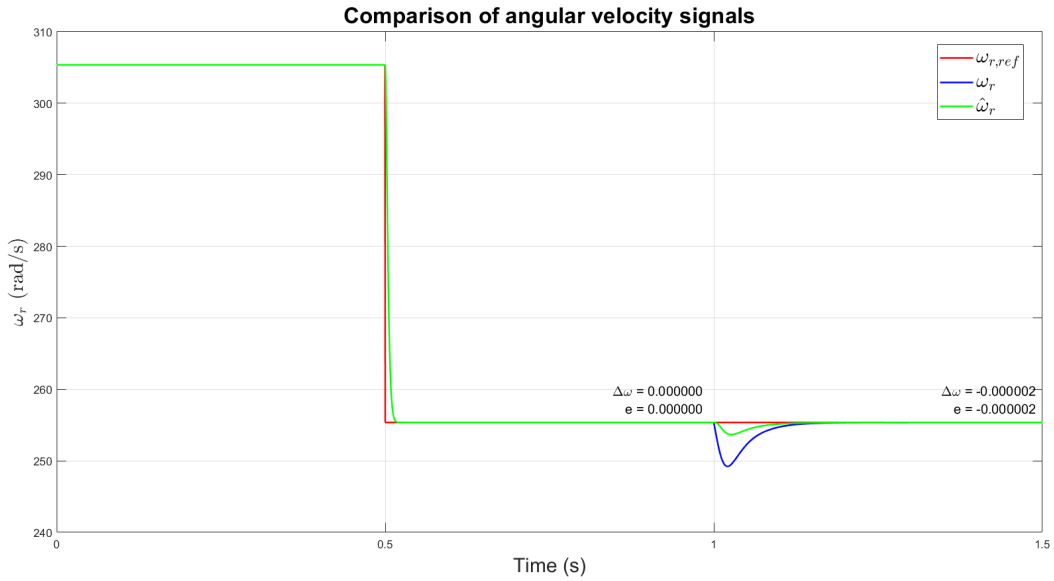


Figure 6: Performance of the output feedback controller using the observer.

The results indicate that when both the system and the observer are initialized at equilibrium, stability is preserved as expected. During the second interval, the system accurately tracks the reference trajectory with an overdamped response, reaching the desired speed in approximately 0.02 s. The observer, when provided with appropriate initial conditions, precisely estimates the real system states without noticeable deviation.

A notable difference from the ideal case, where only a minor disturbance was observed, is the presence of a more pronounced perturbation following the variation in external torque. However, this disturbance remains shorter in duration and amplitude compared to the case of the controller based on singular perturbations, specially in terms of the observer. As

the observer progressively refines its estimation, the motor speed also converges toward the reference speed, without the need to wait for the estimation to fully stabilize beforehand. Once this process is complete, equilibrium is achieved within approximately 0.15 s.

Regarding absolute errors, no significant deviations were observed during the second interval, with absolute errors remaining within six decimal places. In the final interval, an estimation error on the order of 10^{-6} is observed, which is attributed to numerical inaccuracies within the simulation.

5.4.2 Sensitivity results to modeling errors

The simulation conditions and time intervals remain the same as those presented in 5.4.1. In this section, we analyze the system's response under model uncertainties by considering two cases: one with a 10% error and another with a 25% error. These errors are directly applied to the four parameters that define the system, as specified in 6.

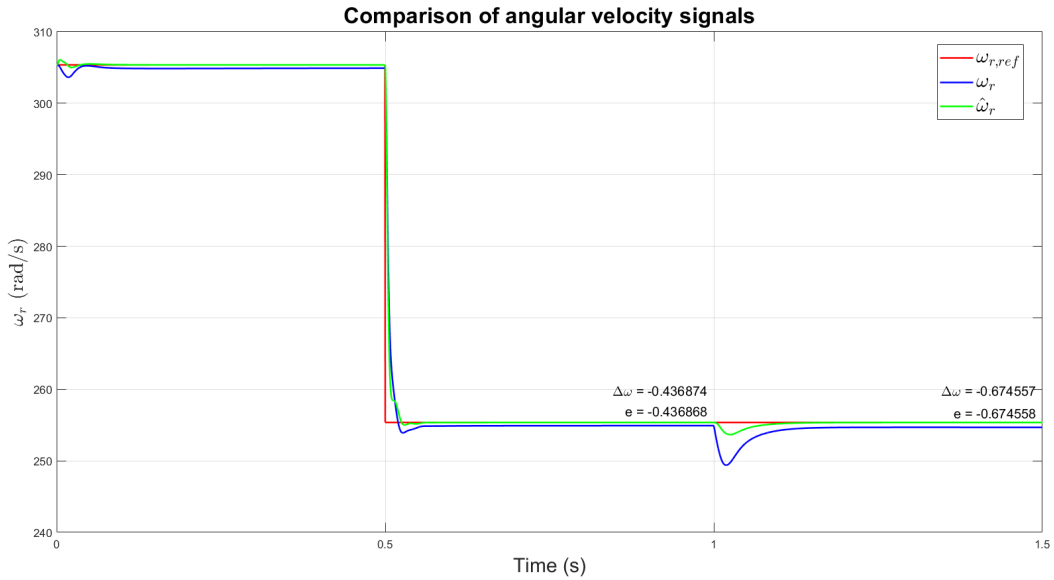


Figure 7: Controller response with a 10% modeling error in the motor parameters

In these two cases, similar to the previous control approach, the system initially destabilizes after starting from equilibrium. However, unlike the previous results, it is now the actual motor speed that exhibits a larger perturbation amplitude. Nevertheless, the stabilization occurs more rapidly, with the system settling after approximately 0.07 s in the first case and 0.1 s in the second case, where the modeling error is greater. Upon stabilization, the estimated value successfully tracks the reference, whereas the actual speed exhibits a steady-state error that increases with the modeling inaccuracy.

During the reference speed change, the system demonstrates a fast response characterized by an underdamped behavior with a settling time of approximately 0.07 s in both cases. However, the second case, with higher modeling errors, presents a significantly larger overshoot. The system ultimately stabilizes with a steady-state error of 0.4 rad/s in the first case and 0.6 rad/s in the second case.

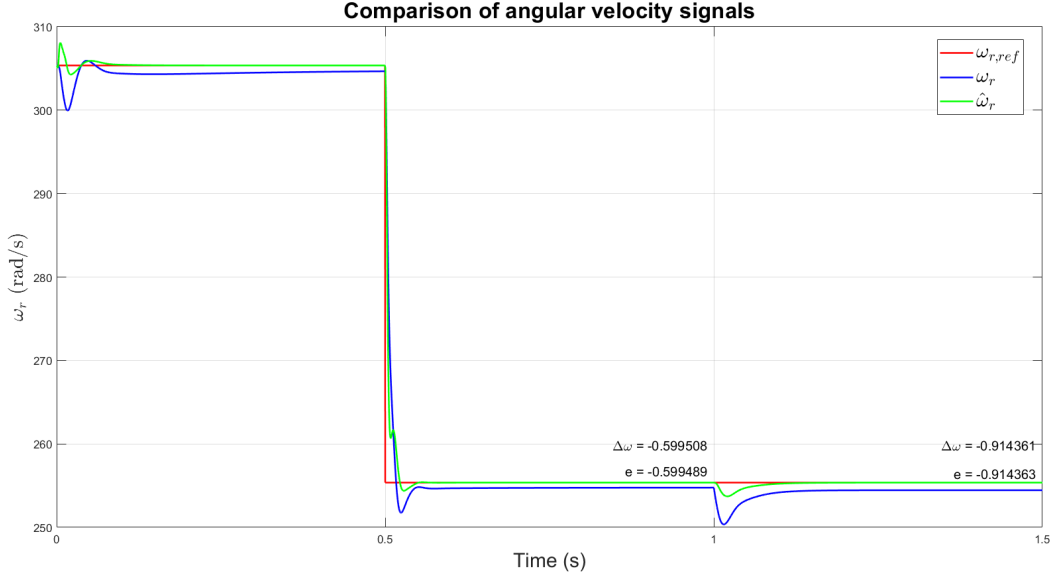


Figure 8: Controller response with a 25% modeling error in the motor parameters

Finally, during the external torque variation, the observer's behavior remains consistent with the ideal case, experiencing only a minor perturbation. However, the actual motor speed exhibits a more pronounced deviation, along with a persistent estimation error. After approximately 0.15 s, the system restabilizes, though a steady-state error remains due to the estimation inaccuracies. This error reaches 0.67 rad/s in the first case and 0.91 rad/s in the second case, which, even in the worst-case scenario, corresponds to only 0.3% of the motor speed.

6 Conclusions

This study provides a comparative analysis of sensorless control strategies for induction motors using singular perturbation-based control and feedback linearization. The evaluation focuses on robustness against modeling errors and estimation inaccuracies, with an emphasis on practical application in industrial environments.

- The singular perturbation-based control leverages the natural time-scale separation in motor dynamics, providing stable performance even in the presence of parameter uncertainties. However, this approach tends to exhibit a longer settling time to reach the reference, up to ten times longer than feedback linearization under the tuning used in this study. This makes it less suitable for applications where fast dynamic responses are required, but it remains effective in uncertain environments where modeling inaccuracies are present, offering better tolerance to such uncertainties.
- Feedback linearization effectively cancels system nonlinearities, enabling precise trajectory tracking and superior transient response under ideal conditions. However, its performance is highly sensitive to parameter variations, making it less robust in environments with significant modeling errors. In cases of considerable discrepancies between

the model and actual system behavior, the performance of this method may degrade significantly.

- While neither control strategy guarantees absolute robustness against modeling errors, the simulation results indicate that steady-state speed deviations are minimal, typically within tenths of RPM. For relatively high-speed applications, these deviations can be considered negligible, making both sensorless control strategies viable for industrial systems where cost-effectiveness and reliability are prioritized over pinpoint accuracy.
- The estimation errors of the flux and velocity subsystems converge exponentially to zero. This result is derived from the proof of Uniform Complete Observability (UCO) and Uniform Complete Controllability (UCC), which are essential properties for ensuring exponential convergence. Through the analysis of exponential convergence, we have demonstrated that the states ψ_s , ω_r , and T_l converge exponentially to their true values.
- The selection of the most suitable control strategy depends on the specific demands of the application. Feedback linearization is ideal for situations requiring high precision and fast dynamics, assuming accurate motor parameters are available. On the other hand, singular perturbation-based control is more appropriate for systems where there are significant modeling uncertainties and the need for absolute precision is less critical.
- Despite their strengths, both control strategies have inherent limitations. Feedback linearization relies heavily on precise motor models, which can limit its applicability in systems with significant uncertainties. Additionally, while both strategies exhibit robustness against moderate modeling errors, a fully robust solution remains an open challenge. Furthermore, this study has been limited to numerical simulations; real-world implementation may introduce additional complexities such as measurement noise, actuator constraints, and unmodeled dynamics.
- Future work should focus on developing adaptive control mechanisms to reduce sensitivity to modeling errors, particularly in feedback linearization-based approaches. Observer-based parameter estimation could also improve robustness by dynamically adjusting control parameters in response to system variations. Experimental validation is essential to assess the practical feasibility of these methods and refine them for real-world applications.

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